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APUL

NEET FLIX

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Electrostatics

1. What will happen when we rub a glass rod with silk cloth?
(1) Some of the electrons from the glass rod are transferred to the silk cloth.
(2) The glass rod gets positively charged and silk cloth gets negatively charged.
(3) New charge is created in the process of rubbing.
(4) Both (1) and (2) are correct.

PEL328

2. Object may acquire an excess or deficiency of charge by :
(1) Electric force (2) Heating
(3) Shaking (4) By rubbing

PEL329

3. A method for charging a conductor without bringing a charged body in contact with it is called :
(1) Magnetization
(2) Electrification
(3) Electrostatic induction
(4) Electromagnetic induction

PEL330

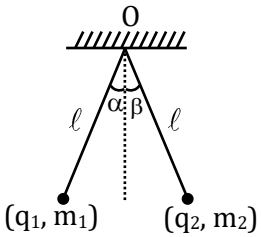
4. Which of the following charges can not be present on an oil drop in Millikan's experiment:
(1) 4.0×10^{-19} C (2) 6.0×10^{-19} C
(3) 10.0×10^{-19} C (4) all of them

PEL331

5. A polythene piece rubbed with wool is found to have a negative charge of 6×10^{-7} C. The number of electrons transferred to polythene from wool is :
(1) 3.75×10^{10} (2) 9.6×10^{10}
(3) 9.6×10^{12} (4) 3.75×10^{12}

PEL332

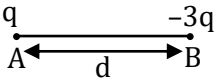
6. Figure shows suspended charge ball system. If $\alpha > \beta$ then at equilibrium which of following may be true :-



(1) $q_1 > q_2$ & $m_1 = m_2$ (2) $q_1 = q_2$ & $m_1 = m_2$
(3) $q_1 > q_2$ & $m_1 < m_2$ (4) $q_1 = q_2$ & $m_1 > m_2$

PEL333

7. Two charges q and –3q are placed fixed on x-axis separated by distance d. Where should a third charge 2q be placed from point A such that it will not experience any force?



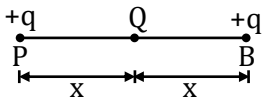
(1) $\frac{d-\sqrt{3}d}{2}$ (2) $\frac{d+\sqrt{3}d}{2}$
(3) $\frac{d+3d}{2}$ (4) $\frac{d-3d}{2}$

PEL334

8. SI unit of permittivity of free space is :
(1) Farad (2) Weber
(3) $C^2 N^{-1} m^{-2}$ (4) $C^2 N^{-1} m^{-1}$

PEL335

9. A charge Q is placed at the centre of the line joining two point charge +q and +q as shown in figure. For system to be in equilibrium the ratio of charges Q and q is :



(1) 4 (2) 1/4 (3) –4 (4) –1/4

PEL336

10. Electric field is maximum due to a charged ring of radius R at a distance from circumference of the ring :
(1) $\frac{R}{\sqrt{2}}$ (2) $\sqrt{\frac{3}{2}}R$ (3) $\frac{\sqrt{3}}{2}R$ (4) $\sqrt{\frac{3}{2}R}$

PEL337

11. Which of the following is true inside a conducting sphere with a net positive charge that is insulated from the ground?
(1) The electric field and the electric potential are zero.
(2) The electric field is not zero but electric potential is zero.
(3) The electric field and electric potential are non zero and equal.
(4) The electric field is zero, and the electric potential is equal to the electric potential at the surface.

PEL338

12. A force of 2.25 N acts on a charge of 15×10^{-4} C. The intensity of electric field at the point is:

(1) 150 N C^{-1} (2) 15 N C^{-1}

(3) 1500 N C^{-1} (4) 1.5 N C^{-1}

PEL339

13. A circular plane sheet of radius 10 cm is placed in a uniform electric field of 5×10^5 N C^{-1}, making an angle of 60° with the field. The electric flux through the sheet is :

(1) 1.36×10^2 N m^2 C^{-1}

(2) 1.36×10^4 N m^2 C^{-1}

(3) 0.515×10^2 N m^2 C^{-1}

(4) 0.515×10^4 N m^2 C^{-1}

PEL340

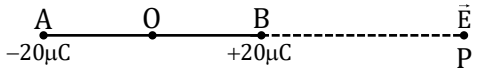
14. A uniform electric field $E = 2 \times 10^3$ N C^{-1} is acting along the positive x-axis. The flux of this field through a square of 10 cm on a side whose plane is parallel to the yz plane is :

(1) 20 N C^{-1} m^2 (2) 30 NC^{-1} m^2

(3) 10 N C^{-1} m^2 (4) 40 N C^{-1} m^2

PEL341

15. Two charges ± 20 μC are placed 10 mm apart. The electric field at point P, on the axis of the dipole 10 cm away from its centre O on the side of the positive charge is :



(1) 8.6×10^9 N C^{-1} (2) 4.1×10^6 N C^{-1}

(3) 3.6×10^6 N C^{-1} (4) 4.6×10^5 N C^{-1}

PEL342

16. A dipole of electric dipole moment p is placed in a uniform electric field of strength E. If θ is the angle between positive directions of p and E, then the potential energy of the electric dipole is largest when θ is :

(1) $\frac{\pi}{4}$ (2) $\frac{\pi}{2}$ (3) π (4) zero

PEL343

17. An electric dipole is placed at an angle of 30° with an electric field of intensity 2×10^5 N C^{-1}. It experiences a torque equal to 4 N m. The charge on the dipole if the dipole length is 2 cm is :

(1) 8 mC (2) 4 mC (3) 6 mC (4) 2 mC

PEL344

18. Match the following and find the correct option.

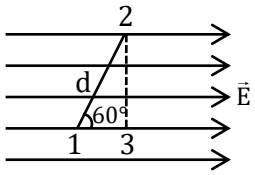
	Column-I		Column-II
(A)	Linear charge density	(p)	$\frac{\text{Charge}}{\text{Volume}}$
(B)	Surface charge density	(q)	$\frac{\text{Charge}}{\text{Length}}$
(C)	Volume charge density	(r)	$\frac{\text{Charge}}{\text{Area}}$

(1) A-q, B-r, C-p (2) A-p, B-r, C-q

(3) A-r, B-p, C-q (4) A-r, B-q, C-p

PEL345

19. The electric field $\vec{E}$ is constant in both magnitude and direction. Consider a path of length d at an angle θ = 60° with respect to field lines as shown in figure. The potential difference between points 1 and 2 is :-

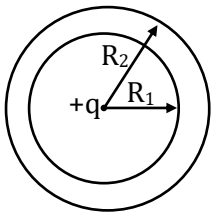


(1) $\frac{E}{d\sin 60°}$ (2) Ed cos 60°

(3) $\frac{Ed}{\cos 60°}$ (4) $\frac{E}{d}\sin 60°$

PEL346

20. A metallic spherical shell has an inner radius R_1 and outer radius R_2. A charge is placed at the centre of the spherical cavity. The surface charge density on the inner surface is :

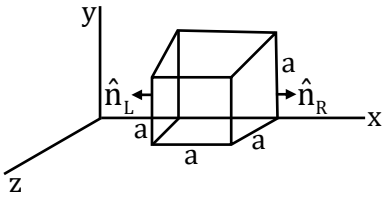


(1) $\frac{q}{4\pi R_1^2}$ (2) $\frac{-q}{4\pi R_1^2}$

(3) $\frac{q^2}{4\pi R_2^2}$ (4) $\frac{q}{4\pi R_2^2}$

PEL347

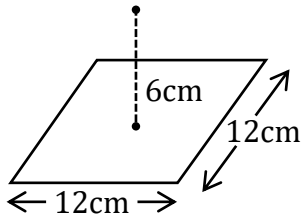
21. The electric field components in the given figure are $E_x = \alpha x^{1/2}$, $E_y = E_z = 0$ in which $\alpha = 800$ N C^{-1} $m^{-1/2}$ if net flux through the cube is 1.05 N m^2 C^{-1}. The charge within the cube is (assume a = 0.1 m)



(1) 9.27×10^{-12} C (2) 9.27×10^{-12} C
(3) 6.97×10^{-12} C (4) 6.97×10^{12} C

PEL348

22. A point charge +20 μC is at a distance 6 cm directly above the centre of a square of side 12 cm as shown in figure. The magnitude of electric flux through the square is :



(1) 2.5×10^6 N m^2 C^{-1}
(2) 3.8×10^5 N m^2 C^{-1}
(3) 4.2×10^5 N m^2 C^{-1}
(4) 2.9×10^6 N m^2 C^{-1}

PEL349

23. If 10 billion electrons move out of a body to Another body every second, How much time is required to get a total charge of one C on the other body?

(1) Two years (2) Two years
(3) 20 years (4) 200 years

PEL350

24. A small sphere of radius r_1 and charge q_1 is enclosed by a spherical shell of radius r_2 and charge q_2. If q_1 is positive & sphere and shell are connected through wire then :-

(1) Charge will flow from the sphere to shell.
(2) Charge will flow from the shell to sphere.
(3) Charge flow will depend on the magnitude q_2.
(4) Charge flow will depend on the magnitude of charge q_1.

PEL351

25. Two points A and B are located in diametrically opposite directions of a point charge of +2 μC at distances 2 m and 1 m respectively from it. The potential difference between A and B is :

(1) 3×10^3 V (2) 6×10^4 V
(3) -9×10^3 V (4) -3×10^3 V

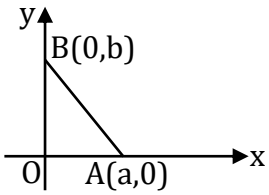
PEL352

26. The electric potential at a point in free space due to a charge Q coulomb is $Q \times 10^{11}$ V. The electric field at that point is :

(1) $12\pi\varepsilon_0 Q \times 10^{22}$ V m^{-1}
(2) $4\pi\varepsilon_0 Q \times 10^{22}$ V m^{-1}
(3) $12\pi\varepsilon_0 Q \times 10^{20}$ V m^{-1}
(4) $4\pi\varepsilon_0 Q \times 10^{20}$ V m^{-1}

PEL353

27. A charge +q is placed at the origin O of x-y axes as shown in the figure. The work done in taking a charge Q from A to B along the straight line AB is



(1) $\frac{qQ}{4\pi\varepsilon_0}\left(\frac{a-b}{ab}\right)$ (2) $\frac{qQ}{4\pi\varepsilon_0}\left(\frac{b-a}{ab}\right)$
(3) $\frac{qQ}{4\pi\varepsilon_0}\left(\frac{b}{a^2}-\frac{1}{b}\right)$ (4) $\frac{qQ}{4\pi\varepsilon_0}\left(\frac{a}{b^2}-\frac{1}{b}\right)$

PEL354

28. An electric dipole of length 20 cm having $\pm 3 \times 10^{-3}$ C charge placed at 60° with respect to a uniform electric field experiences a torque of magnitude 6 N m.

The potential energy of the dipole is :

(1) $-2\sqrt{3}$J (2) $5\sqrt{3}$J
(3) $-3\sqrt{2}$J (4) $3\sqrt{5}$J

PEL355

29. An electric dipole is placed at the centre of a hollow conducting sphere. Which of the following is correct?

(1) Electric field is zero at every point of the sphere.

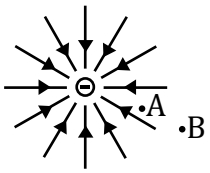
(2) Electric field is not zero anywhere on the sphere.

(3) The flux of electric field is not zero through the sphere

(4) All of these

PEL356

30. Figure shows the field lines of a point negative charge. In going from B to A, the kinetic energy of a small negative charge will :



(1) Increase

(2) Decrease

(3) Remains constant

(4) Data insufficient

PEL357

31. Two thin wire rings each having a radius R are placed at a distance d apart with their axes coinciding. The charge on the two rings are +q and –q. The potential difference between the centres of the two rings is :

(1) $\frac{q}{4\pi\varepsilon_0}\left[\frac{1}{R}-\frac{1}{\sqrt{R^2+d^2}}\right]$

(2) zero

(3) $\frac{q}{2\pi\varepsilon_0}\left[\frac{1}{R}-\frac{1}{\sqrt{R^2+d^2}}\right]$

(4) $\frac{qR}{4\pi\varepsilon_0 d^2}$

PEL358

32. Which of the following statement is true about the relation between electric field and potential?

(1) Electric field is in the direction in which the potential decreases steepest.

(2) Magnitude of electric field is given by the change in the magnitude of potential per unit displacement normal to the equipotential surface at that point.

(3) In the region of strong electric field, equipotential surfaces are far apart.

(4) Both the statements (1) and (2) are correct.

PEL359

33. The top of the atmosphere is about at 400 kV with respect to the surface of earth, corresponding to an electric field that decreases with altitude. Near the surface of earth the field is about 100 V m^{-1}, but still don't get an electric shock, as we set out of our houses in open because (assume the house is free from electric field)

(1) Our body is a perfect insulator

(2) Our body and ground form an equipotential surface.

(3) The original equipotential surfaces of open air remain same.

(4) None of these

PEL360

34. The work done in carrying a charge q once round a circle of radius a with a charge Q at its centre is :

(1) $\frac{qQ}{4\pi\varepsilon_0 a}$ (2) $\frac{qQ}{4\pi\varepsilon_0 a^2}$

(3) $\frac{q}{4\pi\varepsilon_0 a}$ (4) zero

PEL361

35. Four equal charges q each are placed at four corners of a square of side a each. Work done in carrying a charge –q from its centre to infinity is :

(1) zero (2) $\frac{\sqrt{2}q^2}{\pi\varepsilon_0 a}$

(3) $\frac{\sqrt{2}q}{\pi\varepsilon_0 a}$ (4) $\frac{q^2}{\pi\varepsilon_0 a}$

PEL362

36. Match the column :

I		II	
(A)	Electric charge	(P)	$[M^0L^0T^1A^1]$
(B)	Electric field	(Q)	$[M^1L^3T^{-3}A^{-1}]$
(C)	Electric flux	(R)	$[M^1L^1T^{-3}A^{-1}]$
(D)	Electric potential	(S)	$[M^1L^2T^{-3}A^{-1}]$

(1) A→P, B→R, C→Q, D→S
(2) A→P, B→Q, C→R, D→S
(3) A→P, B→S, C→Q, D→R
(4) A→R, B→S, C→Q, D→P

PEL363

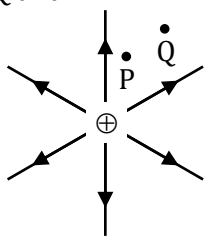
37. Potential due to dipole having dipole moment $\vec{p}$ and position vector of interested point which is at large distance away is $\vec{r}$ the

(1) $V=\frac{1}{4\pi\varepsilon_0}\left|\frac{\vec{p}\times\vec{r}}{r^2}\right|$ (2) $V=\frac{1}{4\pi\varepsilon_0}\frac{\vec{p}\cdot\vec{r}}{r^2}$

(3) $V=\frac{1}{4\pi\varepsilon_0}\frac{\vec{p}\cdot\hat{r}}{r^2}$ (4) $V=\frac{1}{4\pi\varepsilon}\frac{\hat{p}\cdot\vec{r}}{r^2}$

PEL364

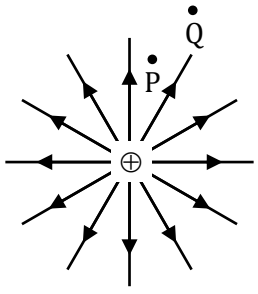
38. Give the sign of the potential energy difference of a small negative charge between the points Q and P



(1) –ve
(2) +ve
(3) zero
(4) may be +ve or –ve

PEL365

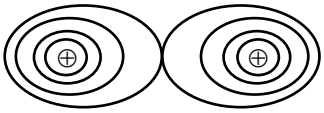
39. Give the sign of the work done by the field in moving a small positive charge from Q to P.

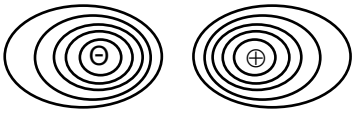


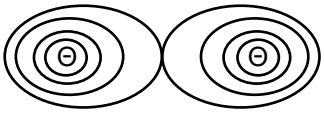
(1) –ve (2) +ve
(3) zero (4) may be +ve or –ve

PEL366

40. Find the correct options regarding Equipotential surface for given charge distribution :-

(1) 

(2) 

(3) 

(4) All of these

PEL367

41. The degree of divergence between the leaves of gold leaf electroscope is an indicator of: -
(1) Amount of charge
(2) Amount of electric field
(3) Amount of potential
(4) Amount of mass

PEL368

42. When we rub a glass rod with silk some of the
(1) Electrons from the rod are transferred to the silk cloth
(2) Protons from the rod are transferred to the silk cloth
(3) Electrons from the silk cloth are transferred to the rod
(4) Protons from the silk cloth are transferred to the rod

PEL369

43. At which level the quantization of charge has no practical consequences and can be ignored?
(1) Microscopic level
(2) Macroscopic level
(3) Both at micro and microscopic level
(4) Can't say anything

PEL370

44. **Assertion(A) : -** Electrostatic field lines do not form any closed loops.

Reason (R): - The conservative nature of electric field.

(1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.

(2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.

(3) **(A)** is true but **(R)** is false.

(4) **(A)** is false but **(R)** is true.

PEL371

45. A dipole is referred to as a point dipole when

(1) The dipole size approaches 0

(2) The charge approaches infinity

(3) Dipole moment is finite

(4) All conditions must have

PEL372

46. Any violation of gauss's law will indicate departure from the

(1) inverse is square law

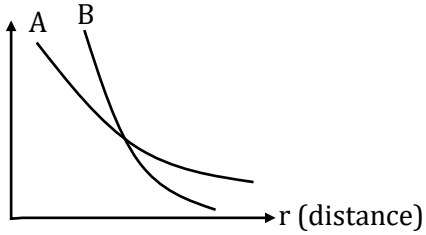
(2) charge conservation law

(3) mass conservation law

(4) momentum conservation law

PEL373

47. Find the correct option regarding point charge.



(1) A → field B → potential

(2) A → potential B → field

(3) None of these have potential curves

(4) None of these have field curves

PEL374

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	4	4	3	4	4	3	2	3	4	2	4	3	2	1	3
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	3	4	1	2	2	1	2	3	1	3	2	1	1	2	2
Question	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Answer	3	4	2	4	2	1	3	2	1	4	1	1	2	1	4
Question	46	47													
Answer	1	1													

Current Electricity

1. Two identical cells are connected in (a) series (b) parallel then maximum power transferred to the load is :-

(1) Same in both cases

(2) Different in both cases

(3) May or may not remain same depending upon polarity of the cell connected

(4) None of these

PCE324

2. Two cells of unequal emf E_1 and E_2 having internal resistance r_1 and r_2 are connected as shown in figure: -



(1) Potential difference across any of the cell cannot be zero

(2) Potential difference across cell of smaller EMF may be zero

(3) Potential difference across cell of larger EMF may be zero

(4) Potential difference can be zero only if EMF of both cell are same

PCE325

3. Conductivity of electrolyte is very low as compared to metal at room temperature because :-

(1) Charge of each ion of electrolyte is very large as compare to charge of each free e^{-1}

(2) Free ion density in electrolyte is much smaller as compared to e^{-1} density of metal

(3) Free ion density in electrolyte is very large as compared to e^{-1} density of metal

(4) None of these

PCE326

4. The standard resistance coils are made of manganin because :-

(1) Manganin wire is very sensitive to change in temperature

(2) For manganin the temperature coefficient of resistance varies hyperbolically for large range of temperature

(3) For Manganin, the temperature coefficient of resistance is almost zero

(4) None of these

PCE327

5. Electrical conductivity of a typical insulator is about 10^{-22} times that of metal while heat conductivity is about 10^{-3} because :-

(1) In electrical conduction only free e^- take part while in thermal conduction both lattice and free e^- contribute

(2) During electrical conduction no. of free e^- are considerably lesser as compare to the number during thermal conduction

(3) During electrical conduction no. of collision of free e^- are extremely large whereas during thermal conductor no. of collision are small

(4) None of these

PCE328

6. A compartment of a train is lit by 20 identical bulb in series. one of the bulbs fuses and is removed from the chain. The remaining bulbs now light the room. The light in the compartment will :-

(1) Increases

(2) Decreases

(3) Neither increases nor decreases

(4) None of these

PCE329

7. Kirchoff's voltage law in a circuit indicates :-

(1) Electric field inside current carrying conductor is non zero

(2) Electric field outside current carrying conductor is always zero

(3) Electrostatic field is conservative field

(4) None of the above

PCE330

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8. A piece of copper and other of germanium (semiconductor) are cooled from room temperature to 100 K :-
(1) Conductivity of copper increases and that of germanium decrease
(2) Conductivity of copper decreases and that of germanium increases
(3) Conductivity of copper as well as germanium both decreases
(4) Conductivity of copper as well as germanium both increases

PCE331

9. A constant potential difference is maintained across a conductor. If the temperature of conductor is increased, drift speed of electron will:-
(1) Increase (2) Decrease
(3) Remain same (4) Can't say

PCE332

10. On bending a conducting wire its :-
(1) Resistance will increase because drift velocity of electron in bent wire decreases
(2) Resistance will decrease because drift velocity of electron in bent wire increases
(3) Resistance will remain same because drift velocity of electron in bent wire will remain same
(4) Resistance will remain same but drift velocity of electron in bent wire will decrease

PCE333

11. In which substance, positive and negative charges both can move?
(1) non-electrolytic solution
(2) Electrolytic solution
(3) Both (1) and (2)
(4) Neither (1) nor (2)

PCE334

12. If we consider a mechanism where, the ends of the cylinder are supplied with fresh charges to make up for any charges neutralised by electrons moving inside the conductor, in that case: -
(1) There will be steady electric field in the conductor
(2) There will be continuous current rather than a current for a short period of time
(3) Both (1) and (2)
(4) Neither (1) nor (2)

PCE335

13. When current flows through copper wire, current density J, electric field E and motion of electrons have directions such that: -
(1) J & E in opposite direction
(2) Motion of electrons and E, in opposite direction
(3) J & motion of electrons in same direction
(4) J, E & motion of electrons are in same direction

PCE336

14. High current in solid conductor can be obtained because :-
(1) drift speed is large
(2) charge on electron is large
(3) electron number density is enormous $\sim 10^{29}\ m^{-3}$
(4) None of the above

PCE337

15. Is it possible that any battery has some constant non-zero value of emf but the potential difference between the plates is zero?
(1) Not possible
(2) Yes, if another identical battery is joined in supportive and series
(3) Yes, if another identical battery is joined in opposition and series
(4) Yes, possible, if another similar battery is joined in parallel

PCE338

16. In meter bridge for measurement of resistance the known and the unknown resistances are inter changed. The error so removed is :-
(1) end correction
(2) index error
(3) due to temperature effect
(4) random error

PCE339

17. Which of the following characteristics of electrons determines the current in a conductor?
(1) Drift velocity alone
(2) Thermal velocity alone
(3) Both drift velocity and thermal velocity
(4) Neither drift nor thermal velocity

PCE340

18. Kirchhoff's junction rule is a reflection of :
(1) Conservation of current density vector
(2) Conservation of energy
(3) The fact that the momentum with which a charged particle approaches a junction is unchanged (as a vector) as the charged particle leaves the junction
(4) The fact that there is no accumulation of charges at a junction

PCE341

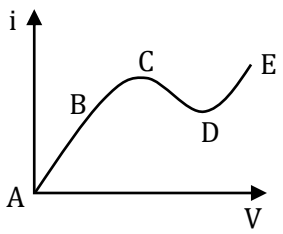
19. The electrical resistance of a conductor depends upon :-
(1) Size of conductor
(2) Temperature of conductor
(3) Geometry of conductor
(4) All of these

PCE342

20. For which of the following dependences of drift velocity V_d on electric field E, is Ohm's law obeyed ?
(1) $V_d \propto E$ (2) $V_d \propto E^2$
(3) $V_d \propto \sqrt{E}$ (4) $V_d \propto \frac{1}{E}$

PCE343

21. From the graph between current i & voltage V shown, identity the portion corresponding to negative resistance



(1) DE (2) CD
(3) BC (4) AB

PCE344

22. An electric heater is connected to the voltage supply. After few seconds, current get its steady value then its initial current will be :-
(1) equal to its steady current
(2) Slightly higher than steady current
(3) Slightly less than its steady current
(4) Zero

PCE345

23. In a wheat stone bridge if the battery and galvanometer are interchanged the deflection in Galvanometer will :
(1) Increase in previous direction
(2) Not change
(3) Change in opposite direction
(4) None of these

PCE346

24. In parallel combination of n-cells we obtain :-
(1) More voltage
(2) More current
(3) Less voltage
(4) Less current

PCE347

25. Match the Column-I with Column-II.

	Column-I		**Column-II**
(A)	Ohm's law is applicable to	(p)	Metals
(B)	Ohm's law is not applicable	(q)	greater resistivity
(C)	Alloys have	(r)	diodes, electrolytes, semiconductors

(1) A-r, B-q, C-p
(2) A-p, B-r, C-q
(3) A-r, B-p, C-q
(4) A-q, B-r, C-p

PCE348

26. Match the column: –

Column-I		Column-II	
(a)	Mobility	(P)	$M^{-1}L^{0}T^{2}A^{1}$
(b)	Resistance	(Q)	$M^{1}L^{2}T^{-3}A^{-2}$
(c)	Current density	(R)	$M^{1}L^{3}T^{-3}A^{-2}$
(d)	Resistivity	(S)	$M^{0}L^{-2}T^{0}A^{1}$

(1) a → P, b → Q, c → S, D → R
(2) a → Q, b → P, c → S, D → R
(3) a → Q, b → S, c → P, D → R
(4) a → P, b → Q, c → R, D → S

PCE349

27. The paths of electron between successive collisions (with the positive ions of the metal) in the absence of electric field will be: -
(1) Straight line (2) Parabolic
(3) Elliptical (4) Curved

PCE350

28. The paths of electron between successive collisions (with the positive ions of the metal) in the presence of electric field will be
(1) Straight line
(2) Curved
(3) Both
(4) No motion

PCE351

29. Percentage errors in measuring resistance from metre bridge device can be minimised by adjusting the balance point near.
(1) Middle of the bridge
(2) Corner of the bridge
(3) 1/4 to the bridge
(4) 3/4 to the bridge

PCE352

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	3	1	2	3	1	1	3	1	2	3	2	3	2	3	3
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	
Answer	1	1	4	4	1	2	2	2	2	2	1	1	2	1	

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Capacitor

1. A conductor with a positive charge -
(1) Is always at positive potential
(2) Is always at zero potential
(3) Is always at negative potential
(4) May be +ve, zero or –ve potential

PCP207

2. A metallic sphere is given some charge. Electric energy is stored.
(1) Only inside the sphere
(2) Only outside the sphere
(3) Both inside and outside
(4) Neither inside nor outside

PCP208

3. The maximum electric field that a dielectric medium of a capacitor can withstand without break down is called its
(1) Polarisation
(2) Capacitance
(3) Dielectric strength
(4) None of these

PCP209

4. If dielectric constant and dielectric strength be denoted by k & x respectively, then a material suitable for use as a dielectric in a capacitor must have
(1) high k & high x (2) high k & low x
(3) low k & high x (4) low k & low x

PCP210

5. A parallel plate capacitor has a uniform electric field E (V m^{-1}) in the space between the plates. If the distance between the plates is d(m) and area of each plate is A (m^2) the energy (Joule) stored in the capacitor, is :

(1) $\frac{1}{2}\in_0 E^2$ (2) $\in_0 EAd$

(3) $\frac{1}{2}\in_0 E^2 Ad$ (4) $E^2Ad/\in_0$

PCP211

6. Two metal spheres of radii R and 2R having same charge density σ, are brought in contact then charge densities becomes :-

(1) $\frac{\sigma}{3},\frac{2\sigma}{3}$ (2) $\frac{2\sigma}{3},\frac{\sigma}{3}$

(3) $\frac{5\sigma}{3},\frac{5\sigma}{6}$ (4) $\frac{5\sigma}{6},\frac{5\sigma}{3}$

PCP212

7. A capacitor of capacitance C_1 is charged to a potential V and then connected in parallel to an uncharged capacitor of capacitance C_2. The final potential difference across each capacitor will be :-

(1) $\frac{C_1V}{C_1+C_2}$ (2) $\frac{C_2V}{C_1+C_2}$

(3) $\left(1+\frac{C_2}{C_1}\right)V$ (4) $\left(1-\frac{C_2}{C_1}\right)V$

PCP213

8. Two identical capacitors are joined in parallel, charged to a potential V, separated and then connected in series, the positive plate of one is connected to the negative of the other. Which of the following is true ?
(1) The charges on the free plated connected together are destroyed.
(2) The energy stored in the system increases.
(3) The potential difference between the free plates is 2 V.
(4) The potential difference remains constant.

PCP214

9. A parallel plate air capacitor has a capacitance C. When it is half filled with a dielectric of dielectric constant 5, the percentage increase in the capacitance will be:-
(1) 400% (2) 66.6%
(3) 33.3% (4) 200%

PCP215

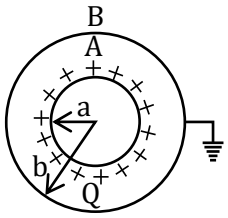
10. Two identical capacitors have the same capacitance C. One of them is charged to potential V_1 and the other to V_2. The negative ends of the capacitors are connected together. When the positive ends are also connected, the decreases in energy of the combine system is :-

(1) $\frac{C}{4}(V_1^2-V_2^2)$ (2) $\frac{C}{4}(V_1^2+V_2^2)$

(3) $\frac{C}{4}(V_1-V_2)^2$ (4) $\frac{C}{4}(V_1+V_2)^2$

PCP216

11. Two spherical conductors A and B of radii a and b (b > a) are placed concentrically in air. A is given a charge +Q while B is earthed. Then the equivalent capacitance of the system is :–

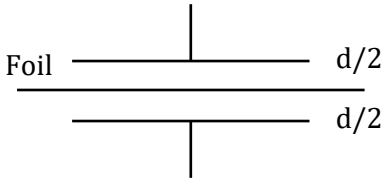


(1) $4\pi\varepsilon_0\left(\frac{ab}{b-a}\right)$ (2) $4\pi\varepsilon_0(a+b)$

(3) $4\pi\varepsilon_0 b$ (4) $4\pi\varepsilon_0\left(\frac{b^2}{b-a}\right)$

PCP217

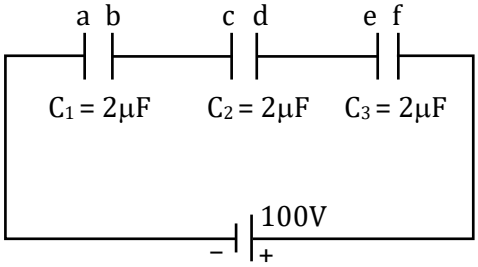
12. A sheet of aluminium foil of negligible thickness is placed between the plates of a capacitor of capacitance C as shown in the figure then capacitance of capacitor becomes



(1) 2C (2) C (3) C/2 (4) zero

PCP218

13. Three condensers C_1, C_2 and C_3 are connected to a 100 volt D.C. source as shown in the figure (Fig.). If the charges stored on the plates of C_1, C_2 and C_3 are q_a, q_b and q_c, q_d and q_e, q_f respectively, then :-



(1) $q_b + q_d + q_f = \frac{100}{9}$ coulomb

(2) $q_b + q_d + q_f = 0$

(3) $q_b = q_d = q_f$

(4) $q_a + q_c + q_e = 50$ Coulomb

PCP219

14. A parallel plate capacitor after charging is kept connected to a battery and the plates are pulled apart with the help of insulating handles. Now which of the following quantities will decrease ?

(1) Charge (2) Capacitance

(3) Energy stored (4) All of these

PCP220

15. An isolated metal sphere of radius R is charged upto potential V. Energy stored in the electric field associated with the charged sphere can be given by

(1) $2\pi\varepsilon_0 RV^2$ (2) $\sqrt{2}\pi\varepsilon_0 RV^2$

(3) $4\pi\varepsilon_0 RV^2$ (4) $\pi\varepsilon_0 RV^2$

PCP221

16. A capacitor of capacitance 2μF is charged by a battery. The battery is removed and another uncharged capacitor of capacitance 6μF is connected is parallel. The total electrostatic energy of resulting system :-

(1) Decreases

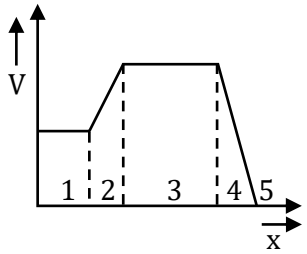
(2) Remains the same

(3) Increases

(4) None of these

PCP222

17. The figure gives the electric potential V as a function of distance through five regions on x-axis. Which of the following is true for the electric field E in these regions :-



(1) $E_1 > E_2 > E_3 > E_4 > E_5$

(2) $E_1 = E_3 = E_5$ and $E_2 < E_4$

(3) $E_2 = E_4 = E_5$ and $E_1 < E_3$

(4) $E_1 < E_2 < E_3 < E_4 < E_5$

PCP223

18. A 2μF condenser is charged upto 200V and then battery is removed. On combining this with another uncharged condenser in parallel, the potential differences between two plates are found to be 40V. The capacity of second condenser is:-

(1) 2μF (2) 4μF (3) 8μF (4) 16μF

PCP224

19. A capacitor is connected with a cell if distance between plates is halved then force of attraction between the plates :-

(1) Remains same (2) Becomes 8 times

(3) Becomes 4 times (4) Becomes 2 times

PCP225

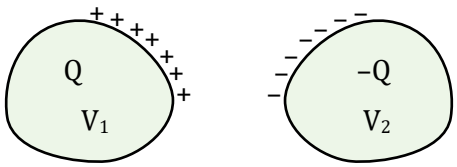
20. A condenser of capacity C_1 is charged to a potential V_0. The electrostatic energy stored in it is U_0. It is connected to another uncharged capacitor C_2 in parallel. The energy dissipated in the process is :

(1) $\frac{C_2}{C_1+C_2}U_0$ (2) $\frac{C_1}{C_1+C_2}U_0$

(3) $\left(\frac{C_1-C_2}{C_1+C_2}\right)U_0$ (4) $\frac{C_1C_2}{2(C_1+C_2)}U_0$

PCP226

21.



Conductor 1 Conductor 2

If a system of two conductors separated by an insulator forms a capacitor, then in the formula of Q = CV, V indicates

(1) $V_1 - (-V_2)$ (2) $V_1 - V_2$

(3) $\frac{V_1V_2}{V_1+V_2}$ (4) $\frac{(V_1+V_2)}{2}$

PCP227

22. Meaning of "fringing of the field" term in reference of capacitors is

(1) bending of field lines from the centre of plates of capacitors

(2) bending of field lines from the edges of plates of capacitors

(3) bending of field lines from the centre & edges of plates of capacitors

(4) bending of field lines from only above of centre of plates of capacitors

PCP228

23. Match the column: -

Column-I		Column-II	
(a)	Electric susceptibility	(P)	$M^{-1}L^{-2}T^{4}A^{2}$
(b)	Dielectric strength	(Q)	$M^{0}L^{0}T^{0}$
(c)	Capacitance	(R)	$M^{1}L^{-1}T^{-2}$
(d)	Electric potential energy density	(S)	$M^{1}L^{1}T^{-3}A^{-1}$

(1) a → Q, b → S, c → P, d → R

(2) a → S, b → Q, c → P, d → R

(3) a → Q, b → S, c → R, d → P

(4) a → Q, b → P, c → S, d → R

PCP229

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	4	2	3	1	3	3	1	3	2	3	1	2	3	4	1
Question	16	17	18	19	20	21	22	23							
Answer	1	2	3	3	1	2	2	1							

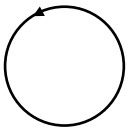
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Magnetic Effect of Current and Magnetism

1. A vertical wire carries a current upwards. The magnetic field at a point due north of the wire is directed
(1) upward (2) due south
(3) due west (4) due east
PME381

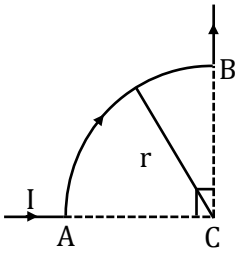
2. Pick out the Wrong statement :
(1) The gain in the K.E. of the electron moving at right angles to the magnetic field is zero.
(2) When an electron is shot at right angles to the electric field, it traces a parabolic path.
(3) An electron moving in the direction of the electric field gains K.E.
(4) An electron at rest experiences no force in uniform and constant magnetic field
PME382

3. A field line is shown in fig. This field cannot represent:-
(1) Magnetic field
(2) Electrostatic field
(3) Induced electric field
(4) All of these



PME383

4. A wire carrying a current I is shaped as shown. Section AB is a quarter circle of radius r. The magnetic field at C is directed



(1) along the bisector of the angle ACB, away from AB
(2) along the bisector of the angle ACB, towards AB
(3) perpendicular to the plane of the paper, directed into the paper
(4) at an angle $\frac{\pi}{4}$ to the plane of the paper
PME384

5. If E and B denote electric and magnetic fields respectively, which of the following is dimensionless?
(1) $\sqrt{\mu_0\varepsilon_0}\frac{E}{B}$ (2) $\mu_0\varepsilon_0\frac{E}{B}$
(3) $\mu_0\varepsilon_0\left(\frac{B}{E}\right)^2$ (4) $\frac{E}{\varepsilon_0}\times\frac{\mu_0}{B}$
PME385

6. Dia-magnetic property of material can be explained by :-
(1) Lenz's law (2) Faraday law
(3) Ampere's law (4) Gauss's law
PME386

7. The magnetic moment produced in a substance of 1 gm is 6×10^{-7} A × m^2. If its density is 5 gm/cm^3, then the intensity of magnetisation in A/m will be: -
(1) 8.3×10^6 (2) 3.0
(3) 1.2×10^{-7} (4) 3×10^{-6}
PME387

8. Superconducting material is :-
(1) Diamagnetic
(2) Perfect diamagnetic
(3) Paramagnetic
(4) Ferromagnetic
PME388

9. When a superconductor is placed near a bar magnet then :-
(1) It repels the magnet
(2) It attracts the magnet
(3) Neither repels nor attracted
(4) Some time repels and some time attracts
PME389

10. A ferromagnetic material is placed in an external magnetic field. The magnetic domains:
(1) increase in size
(2) decrease in size
(3) may increase or decrease in size
(4) have no relation with the field
PME390

11. If magnetic monopoles existed, how would the Gauss's law of magnetism be modified? (Here q_m is the monopole magnetic charge enclosed by surface S)
(1) $\oint\vec{B}.d\vec{S}=\frac{q_m}{\mu_0}$ (2) $\oint\vec{B}.d\vec{S}=q_m$
(3) $\oint\vec{B}.d\vec{S}=\mu_0 q_m$ (4) $\oint\vec{B}.d\vec{S}=\mu_0 q_m^2$
PME391

12. An irregular closed loop carrying a current has a shape such that the entire loop cannot lie in a single plane. If this is placed in a uniform magnetic field, the force acting on the loop :
(1) must be zero
(2) can never be zero
(3) may be zero
(4) will be zero only for one particular direction of the magnetic field

PME392

13. Universal property of all substance is: -
(1) Diamagnetism (2) Paramagnetism
(3) Ferromagnetism (4) All of above

PME393

14. Path of a charged particle moving in a uniform magnetic field cannot be a :
(1) Straight line (2) Circle
(3) Parabola (4) Helix

PME394

15. Material suitable for making electromagnet should have :-
(1) High retentivity and High coercivity
(2) Low retentivity and Low coercivity
(3) High retentivity and Low coercivity
(4) Low retentivity and High coercivity

PME395

16. Magnetic field lines represent the direction :-
(1) Along which a small magnetised needle aligns
(2) Along which moving charge particle experiences a force
(3) Both (1) and (2)
(4) None of (1) and (2)

PME396

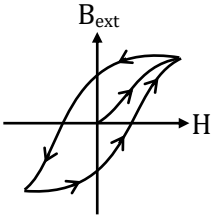
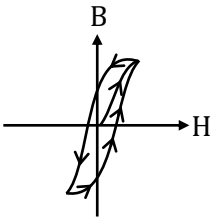
17. For paramagnetic materials :-
(1) χ is positive at all temperature
(2) χ is negative at all temperature
(3) χ may be positive or negative
(4) χ does not depend on temperature

PME397

18. Which of the following statements are correct?
(A) Magnetic field lines can be entirely confined within the core of a toroid
(B) Magnetic field lines can be entirely confined within the core of a solenoid
(C) a bar magnet exert a torque on itself due to its own field.
(D) a system can have magnetic moment even though its net charge is zero.
(1) A, B, D (2) only A, D
(3) A, C (4) A, B, C, D

PME398

19. For the given uses select the correct magnetic material :-

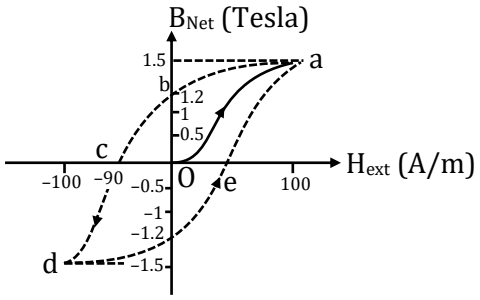
	Column-I		Column-II
(p)	Electromagnet	(A)	
(q)	Transformer	(B)	
(r)	Permanent magnet used in generator		

Codes :

	p	q	r
(1)	A	B	B
(2)	B	B	A
(3)	B	A	B
(4)	A	A	D

PME399

20. The hysteresis curve for a material is shown in figure. Then for the material retentivity, coercivity and saturation magnetic field will be respectively:-



(1) 90 A/m, 1.2 T, 1.5 T
(2) 1.5T, 90 A/m, 1.2 T
(3) 1.2T, 90 A/m, 1.5 T
(4) 90 A/m, 1.5 T, 1.2 T

PME400

21. Select the incorrect statement :-

(1) When a bar magnet of dipole moment $\vec{M}$ is placed in a uniform magnetic field $\vec{B}$ the torque on it is $\vec{M}\times\vec{B}$

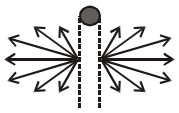
(2) Gauss's law for magnetism states that the net magnetic flux through any closed surface is zero

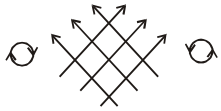
(3) For diamagnetic materials magnetic susceptibility is positive whereas for paramagnetic materials is negative

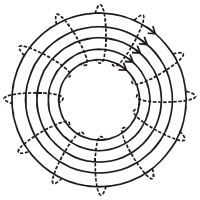
(4) Magnetisation of a paramagnetic material is inversely proportional to the absolute temperature T.

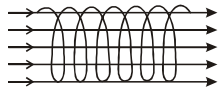
PME401

22. Many of the diagrams given in figure, show magnetic field lines (thick lines in the figure). Point out which one is/are correct :-

(A) 

(B) 

(C) 

(D) 

(1) both A and D (2) both B and D

(3) both C and D (4) only C

PME402

23. When will the field lines be completely expelled (χ = magnetic susceptibility)

(1) $\chi = -1$ and $\mu_r = 0$

(2) $-1 \le \chi$ and $\mu_r >> 1$

(3) $\chi \le 1$ and $\mu_r << 1$

(4) $\chi = 1$ and $\mu_r = 1$

PME403

24. Which statement is wrong for Lorentz force?

(1) $\vec{F}=\vec{F}_{electric}+\vec{F}_{magnetic}$

(2) The magnetic force is zero if charge is not moving

(3) It depends on $\vec{v}$ & $\vec{B}$ only

(4) Force on a negative charge is opposite to that on a positive charge

PME404

25. Which of the following statement is wrong for electric field & magnetic field.

(1) Both are long range.

(2) The principle of superposition applies to both fields

(3) Both fields are along the displacement vector joining the source and the field point.

(4) The electrostatic field is produced by a scalar source & magnetic field produced by a vector source.

PME405

26. Choose correct statement for magnetic field line due to current carrying coil.

(1) The direction of the field is given by right hand palm rule.

(2) Current carrying not behave as magnetic dipole

(3) Field lines are in the form of closed loops

(4) For 'N' turns, magnetic field $B=\frac{\mu_0 NI}{4R}$

PME406

27. Ampere's circuital law can be derived from.

(1) Gauss's law (2) Coulomb's law

(3) Biot savart law (4) None of them

PME407

28. The magnetic field due to current carrying coil of magnetic moment 'm' on the axis of the coil [x > > R].

(1) $B=\frac{\mu_0}{4\pi}\frac{m}{x^2}$ (2) $B=\frac{\mu_0}{4\pi}\frac{2m}{x^2}$

(3) $B=\frac{\mu_0}{4\pi}\frac{m}{x^3}$ (4) $B=\frac{\mu_0}{4\pi}\frac{2m}{x^3}$

PME408

29. A current carrying circular loop is located in a uniform external magnetic field in which orientation, the flux of the total field [external field + field produced by the loop] is maximum.

(1) When angle between area vector and external magnetic field is 90°

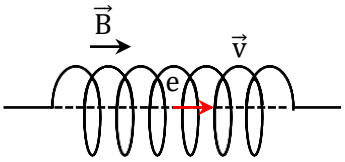
(2) When angle between area vector and external magnetic field is 0°

(3) When angle between area vector and external magnetic field is 150°

(4) When angle between area vector and external magnetic field is 120°

PME409

30. In given diagram, work done to move the electron in direction of magnetic field will be



(1) $W = \vec{F} \cdot \vec{d}$ (2) $W = \vec{M} \cdot \vec{B}$
(3) $W = 0$ (4) $W = qvB\ell$

PME410

31. Two parallel wires are carrying electric currents of equal magnitude and in the same direction they exert.
(1) An attractive force on each other
(2) A repulsive force on each other
(3) No force on each other
(4) A rotational torque on each other

PME411

32. A long hollow tube is carrying an electric current along its length distributed uniformly on its circumference. The magnetic field: -
(1) increases linearly from the axis of the surface
(2) is constant inside the tube
(3) is zero only on the axis
(4) is zero just out-side the tube

PME412

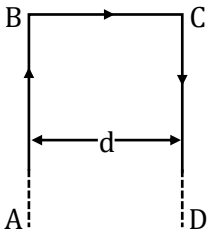
33. For the magnetic field produced by a finitely long current carrying wire. Select wrong statement: -
(1) the lines of force will be concentric circles with centres on the wire
(2) There can be two points in the same plane where magnetic fields are same.
(3) There can be large number of points where the magnetic field is same.
(4) The magnetic field at a point is inversely proportional to the distance of the point from the wire.

PME413

34. The ratio of the energy density of magnetic field in the middle region of a solenoid to that in end region is: -
(1) 4 : 1 (2) 1 : 4 (3) 1 : 1 (4) 2 : 1

PME414

35. The magnetic field at the mid point of BC has magnitude.



(1) $\frac{\mu_0 I}{8\pi d}$ (2) $\frac{\mu_0 I}{4\pi d}$ (3) $\frac{\mu_0 I}{2\pi d}$ (4) $\frac{\mu_0 I}{\pi d}$

PME415

36. Magnetic field at the centre of a circular loop of area 'A' is 'B'. The magnetic moment of the loop is: -

(1) $\frac{BA^2}{\mu_0 \pi}$ (2) $\frac{BA\sqrt{A}}{\mu_0}$
(3) $\frac{BA\sqrt{A}}{\mu_0 \pi}$ (4) $\frac{2BA\sqrt{A}}{\mu_0 \sqrt{\pi}}$

PME416

37. Gauss law of magnetism is: -
(1) Net electric flux through any closed surface is zero
(2) Net magnetic flux through any surface is zero
(3) Net electric flux through any closed surface is (Q/ε_0)
(4) Net magnetic flux through any closed surface is zero

PME417

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	3	3	2	3	1	1	2	2	1	3	3	1	1	3	2
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	1	1	2	2	3	3	4	1	3	3	3	3	4	2	3
Question	31	32	33	34	35	36	37								
Answer	1	2	4	1	4	4	4								

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Alternating Current

1. r.m.s. value of A.C. is defined in terms of :-
(1) heating effect of current
(2) chemical effect of current
(3) magnetic effect of current
(4) none of these

PAC216

2. To express AC power in the same from as DC power, a special value of current is defined and used, is called
(1) root mean square current (I_{rms})
(2) effective current
(3) induced current
(4) both (1) and (2)

PAC217

3. A choke is preferred over a resistance for limiting current in AC circuit because
(1) choke is cheap
(2) there is no wastage of power
(3) choke is compact in size
(4) choke is a good absorber of heat

PAC218

4. If the rms current in a 50 Hz AC circuit is 5 A, the value of the current 1/300 s after its value becomes zero is
(1) $5\sqrt{2}$A (2) $5\sqrt{3/2}$ A
(3) (5/6)A (4) $\left(5/\sqrt{2}\right)$A

PAC219

5. Which circuit, cannot have resonance
(1) R – L (2) R – C
(3) L – C (4) Both (1) and (2)

PAC220

6. In a choke coil, the reactance X_L and resistance R are such that :-
(1) $X_L = R$ (2) $X_L >> R$
(3) $X_L << R$ (4) $X_L = \infty$

PAC221

7. The main advantage of preferring use of AC over DC is-
(1) Easy and efficient conversion from one voltage to other
(2) Lower risk to human life
(3) Most electrical devices work on ac
(4) A variable voltage provides lower average value than the mean value

PAC222

8. Current flowing in a pure resistive AC circuit having resistance 2Ω is given by I = 6 sin(2πt)A. Amount of current in a DC circuit having equal resistance producing same amount of heat in 10 minutes is-
(1) 6 A (2) 4.23 A
(3) 8.48 A (4) 3.26 A

PAC223

9. In a pure inductive circuit, when potential difference $v = v_m \sin \omega t$ is applied, then-
(1) Current reaches its maximum value later than the voltage by one fourth of a period
(2) Current reaches its maximum value later than the voltage by half of a period
(3) Current reaches its maximum value earlier than the voltage by one fourth of a period
(4) Current reaches its maximum value earlier than the voltage by half of a period

PAC224

10. In resonance, as sharpness of resonance decreases then :-
a. Maximum current is increased
b. Maximum current is decreased
c. Circuit becomes more selective
d. Circuit becomes less selective
Select the incorrect statements
(1) b, c (2) b, d (3) a, d (4) a, c

PAC225

11. Out of the given cases, power dissipation is maximum in-
(1) LR circuit
(2) CR circuit
(3) LC circuit
(4) Resonant LCR circuit

PAC226

12. The output of a step-down transformer is measured to be 24 V when connected to a 12W light bulb. The value of the peak current is:
(1) $\left(1/\sqrt{2}\right)$A (2) $\sqrt{2}$ A
(3) 2A (4) $2\sqrt{2}$ A

PAC227

13. Ohm's law for resistors can be applied for-
(1) Only DC
(2) Only AC
(3) Both DC and AC equally
(4) For DC as well as for AC but only when circuit is pure resistive

PAC228

14. Average value of $i = i_0 \sin\omega t$ for half cycle may be
(1) zero (2) $+\frac{2i_0}{\pi}$
(3) $-\frac{2i_0}{\pi}$ (4) All of these

PAC229

15. R.M.S. value of $i = i_0\sin\omega t$ for quarter cycle is
(1) i_0 (2) $\frac{i_0}{\sqrt{2}}$
(3) $\frac{i_0}{2\sqrt{2}}$ (4) None of these

PAC230

16. Average value of pure AC for a complete cycle is
(1) zero (2) 1
(3) –1 (4) None of these

PAC231

17. Power factor for AC circuit which does not contain any resistive component is
(1) Unity (2) Zero
(3) Less than zero (4) All of these

PAC232

18. For an R-L-C series circuit
(i) If $X_L > X_C$, voltage is ahead of current
(ii) If $X_L < X_C$, current is ahead of voltage
Choose the correct option.
(1) (i) only (2) (i) & (ii)
(3) (ii) only (4) None of these

PAC233

19. Consider a AC circuit which is purely inductive or purely capacitive. Then average power absorbed from t = 0 to $t = \frac{\pi}{\omega}$
(applied ac voltage $v = v_m\sin\omega t$)
(1) Zero (2) Non zero
(3) Both (1) & (2) (4) None of these

PAC234

20. At half power frequency, reactance of ac circuit is equal to: -
(1) zero (2) 2R
(3) R (4) None of these

PAC235

21. At half power frequency, phase difference in ac circuit is equal to
(1) zero (2) $\frac{\pi}{4}$ (3) $\frac{\pi}{2}$ (4) $\frac{\pi}{6}$

PAC236

22. Sharpness of resonance in ac circuit depends on
(1) Bandwidth
(2) Resonating frequency
(3) Quality factor
(4) All of these

PAC237

23. Metal detector works on the particle of
(1) Resonance in ac circuits
(2) LC oscillations
(3) Conservation of energy
(4) Conservation of charge

PAC238

24. When a voltage measuring device is connected to ac mains, the meter shows the steady input voltage of 220 V. This means
(1) input voltage cannot be ac voltage
(2) maximum input voltage is 220V.
(3) the meter reads not V but $<V^2>$ and is calibrated to read $\sqrt{<V^2>}$
(4) the pointer of the meter is stuck by some mechanical defect.

PAC239

25. To reduce the resonant frequency in an LCR series circuit with a generator
(1) the generator frequency should be reduced
(2) another capacitor should be added in parallel to the first.
(3) the iron core of the inductor should be removed
(4) dielectric in the capacitor should be removed

PAC240

26. A transformer works on the principle of
(1) self induction
(2) electrical inertia
(3) mutual induction
(4) magnetic effect of electric current

PAC241

27. For an ideal step-down transformer, the quantity which is constant for both primary and secondary coils is
(1) Current in coils
(2) Voltage across coils
(3) Resistance of coils
(4) Power in coils

PAC242

28. The core of a transformer is laminated to reduce energy losses due to
(1) Eddy currents
(2) Hysteresis
(3) Resistance in winding
(4) None of these

PAC243

29. A resistor of resistance 100Ω is connected to an AC source E = 12 sin (250πt). The energy dissipated as heat during t = 0 to t = 1ms is
(1) 2.61×10^{-4} J (2) 2.61×10^{-2} J
(3) 4.84×10^{-4} J (4) 4.84×10^{-2} J

PAC244

30. Alternating current in circuit is given by $I = I_0 \sin 2\pi nt$. Then the time taken by the current to rise from zero to RMS value is equal to

(1) $\frac{1}{2n}$ (2) $\frac{1}{4n}$

(3) $\frac{1}{n}$ (4) $\frac{1}{8n}$

PAC245

31. A source of 220V is applied in an AC circuit. The value of resistance is 220Ω. Frequency and inductance are 50Hz and 0.7H, then wattless current is
(1) 0.5A
(2) 0.7A
(3) 1A
(4) None of the above

PAC246

ANSWER KEY

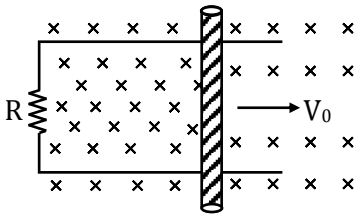
Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	1	4	2	2	4	2	1	2	1	4	4	1	3	1	2
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	1	2	2	1	3	2	4	1	3	2	3	4	1	1	4
Question	31														
Answer	1														

Electromagnetic Induction

1. If a copper plate is allowed to oscillate between (N) pole & (S) pole of a magnet like a simple pendulum, then :-
(1) it will oscillates for ∞ time
(2) it's motion is accelerated
(3) it will not oscillate
(4) it's oscillation is damped

PEI246

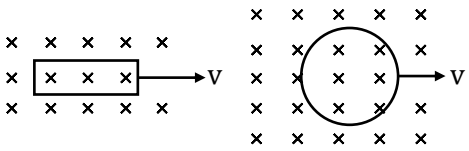
2.



If length of rod '⊥' to magnetic field is released with initial velocity V_0 then :-
(1) It will continue to move with constant velocity
(2) It will accelerate
(3) It will stop after sufficiently long time
(4) None of these

PEI247

3. A rectangular loop and a circular loop are moving out of a uniform magnetic field region in the given figure, to a field free region with a constant velocity v. In which loop do you expect the induced emf to be constant during the passage out of the field region?



(1) Rectangular loop
(2) Circular loop
(3) Both (1) and (2)
(4) Neither (1) nor (2)

PEI248

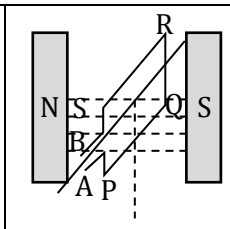
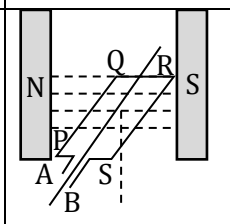
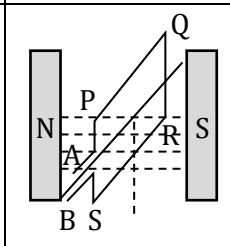
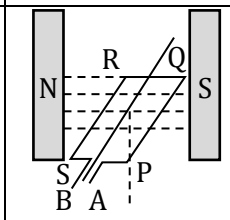
4. A Metallic loop is placed in non uniform steady magnetic field, then
(1) EMF get induced in the loop
(2) No EMF get induced in the loop
(3) An EMF get induced but there will be no current in the loop.
(4) None

PEI249

5. In laminated core, the lamination are separated by an insulating material like
(1) Tar
(2) Lacquer
(3) Canada balsam
(4) Manganin

PEI250

6. Match the column

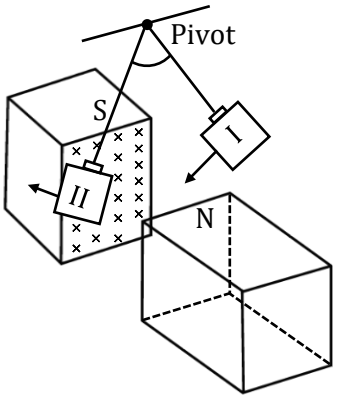
A		P	Area vector parallel or anti parallel to $\vec{B}$
B		Q	$\vec{A} \perp \vec{B}$
C		R	Induce EMF is maximum
D		S	Induce EMF is zero
		T	Flux through coil is maximum
		U	Flux through coil is zero

(1) A-P,R,T , B-Q,R,T , C-P,R,U , D-Q,S,U
(2) A-P,S,T , B-Q,R,U , C-P,S,T , D-Q,R,U
(3) A-Q,R,T , B-P,R,T , C-Q,R,T , D-P,R,T
(4) A-Q,R,U , B-P,S,U , C-Q,R,U , D-P,S,U

PEI251

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7. Find direction of current in position I and position II in the given diagram:



(1) Clockwise in position I and anticlockwise in position II.
(2) Clockwise in position I as well as position II.
(3) Anticlockwise in position I and clockwise in position II.
(4) Anticlockwise in position I as well as position II.

PEI252

8. A closed loop moves normal to the constant electric field between the plates of capacitor such that electric field lines are normal to the plane of the loop. Current will get induced in the loop when:
(1) It is wholly inside the region between the capacitor plates.
(2) It is partially outside the plates of the capacitor.
(3) Electric field lines are parallel to the plane of the loop.
(4) No current is induced in either case.

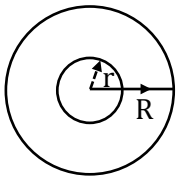
PEI253

9. Which of the following statement is/are true:
a. Charges in motion (current) can exert force/torque on stationary magnet.
b. A Bar magnet in motion (or more generally, a changing magnetic field) can exert force on stationary charge.
c. The fundamental significance of faraday's discovery is that electricity and magnetism are related

(1) Only a (2) a and b
(3) a and c (4) a, b, c

PEI254

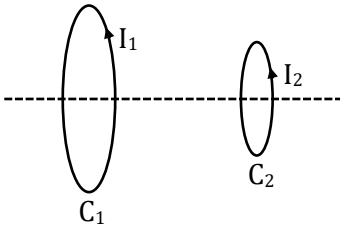
10. Which of the following statement is/ are true for two concentric coplanar coil shown:



Statement 1 : we can find mutually induced EMF and mutual inductance when current in small coil is varying with time
Statement 2 : reciprocity theorem states that $M_{12} = M_{21}$, that is mutual inductance is purely geometrical quantity depending only on size, number of turns, relative position and configuration of two coils but independent of current.
(1) Only statement 1 is correct.
(2) Only statement 2 is correct.
(3) Both statement 1 and statement 2 are correct.
(4) None

PEI255

11. Which of the following statement is true for two nearby coil as shown



A. In the phenomena of mutual induction, self induction of the coil persist
B. Net flux through the coil C can be written as $L_1I_1 + M_{12}I_2$
Where L_1 is self induction of coil 1.
C. Induced EMF in coil 1 can be written as

$$\varepsilon_1 = -L_2 \cdot \frac{dI_1}{dt} - M_{12}\frac{dI_2}{dt}$$

Where L_2 is self induction of coil C_2.
(1) Only A is true
(2) A and B are true
(3) A and C are true
(4) A, B and C are all true

PEI256

12. If a cylindrical magnet is dropped through an aluminium pipe it takes more time to come down then the same cylindrical magnet dropped through a PVC pipe, because:

(1) Magnet attract aluminium pipe and takes longer time whereas there is no attraction with PVC pipe therefore less time to fall down in it.

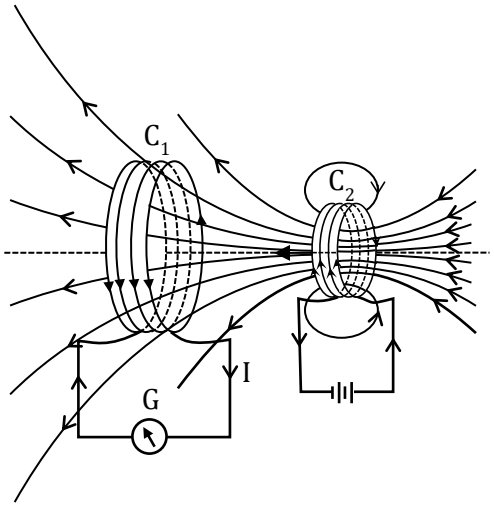
(2) Eddy current are generated in the case of aluminium pipe which opposes the motion of magnet whereas no Eddy current are produced in PVC pipe as it's material is insulator.

(3) Both 1 and 2 are correct.

(4) Both 1 and 2 are incorrect.

PEI257

13. In the given experimental setup which of the following steps can be taken to obtain large deflection of galvanometer



A. Use a rod made of soft iron inside the coil C_2

B. Connect coil C_2 to powerful battery

C. Decrease the resistance of the galvanometer.

D. Move the arrangement rapidly towards C_1 coil.

(1) B, C, D (2) Only D

(3) A, B, C, D (4) B, D

PEI258

14. A square of side L metres lies in the xy-plane in a region, where the magnetic field is given by $B = B_0\left(2\hat{i}+3\hat{j}+4\hat{k}\right)T$, where B_0 is constant.

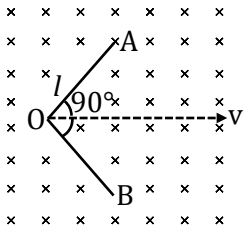
The magnitude of flux passing through the square is: -

(1) $2B_0 L^2$ Wb (2) $3B_0L^2$Wb

(3) $4B_0L^2$Wb (4) $\sqrt{29}\,B_0L^2$Wb

PEI259

15. An ∠AOB made of a conducting wire moves along its bisector through a magnetic field B as suggested by figure. Find the emf induced between A and B, if the magnetic field is perpendicular to the plane of the paper. (where, AO = BO = ℓ)



(1) $\frac{B v \ell}{\sqrt{2}}$ (2) Bv

(3) $\sqrt{2}B\ell v$ (4) $B\ell v$

PEI260

16. Induction furnace make use of

(1) Self induction (2) mutual induction

(3) eddy current (4) none of these

PEI261

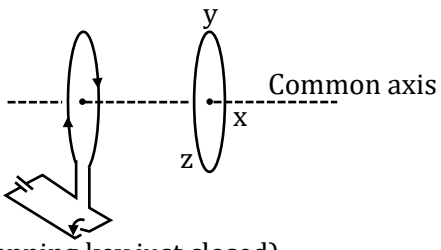
17. In an AC generator, if the plane of coil turns by an angle θ then angle between magnetic field and area vector is

(1) 2θ (2) θ

(3) $\frac{\pi}{2}-\theta$ (4) None of these

PEI262

18. The direction of induced current in the right loop is



(Tapping key just closed)

(1) along common axis

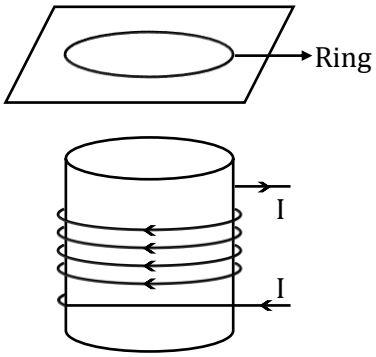
(2) along xzyx

(3) along xyzx

(4) No current

PEI263

19. A metal ring is kept (supported by a card board) on the top of a fixed solenoid which is caring a current I as shown in figure. The centre of ring coincides with the axis of solenoid. If the current is solenoid is switched off, then



(1) magnetic flux linked with metal ring increases.
(2) current induced in metal ring in clockwise direction.
(3) metal ring will not remain on cardboard.
(4) both (1) & (2) are correct.

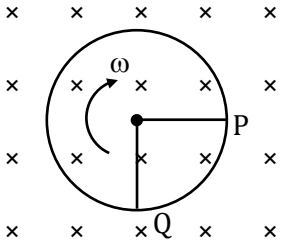
PEI264

20. A solenoid has an inductance of 10 henry and a resistance of 2.5Ω. It is connected to a 10V battery. How long will it take for the magnetic energy to reach $\frac{1}{4}$th of it's maximum value

(1) 0.693s (2) 2.772s
(3) 3.465s (4) 2.142s

PEI265

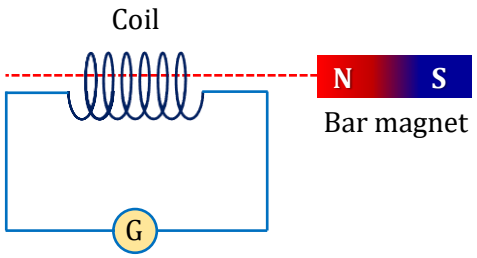
21. A conducting ring of radius r is rolling without slipping with a constant angular velocity ω. If the magnitude field strength is B & is directed into the page then the emf induced across PQ is



(1) $B\omega r^2$ (2) $\frac{B\omega r^2}{2}$
(3) $4B\omega r^2$ (4) $\frac{\pi^2 r^2 B\omega}{8}$

PEI266

22. Figure shows a magnet and a coil. The magnet is moved towards the coil for a short duration of time Δt. Then which of the following induced parameters is independent of Δt.



(1) Induced emf
(2) Induced current
(3) Induced charge flow
(4) Induced heat

PEI267

23. A magnet and a coil are moving towards each other, (there axis lies on same line). If magnet is moving with speed v_1 & coil with speed v_2, emf induced in coil is e_1. If direction of motion of magnet is reversed, emf induced in coil is e_2. Ratio of e_1 & e_2 is

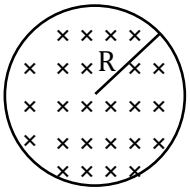
(1) $\frac{v_1 - v_2}{v_1 + v_2}$ (2) $\frac{v_1 + v_2}{v_1 - v_2}$
(3) $\frac{v_1}{v_2}$ (4) $\frac{v_1 + 2v_2}{2v_1 - v_2}$

PEI268

24. The direction of induced electric field is
(1) along magnetic field
(2) along induced current
(3) along the dir. of flow of electrons
(4) None of the above

PEI269

25. A time varying magnetic field is present in a cylindrical region of radius R as shown in figure (cross-sectional view). B is increasing with time then,



$\frac{dB}{dt} = +\alpha\ T/s$

(a) For r < R, the induced electric field is proportional to r.

(b) For r > R, the induced electric field is proportional to $\frac{1}{r}$.

(c) For r = R, the induced electric field is maximum.

(d) If a coaxial "Non conducting" ring of radius $\frac{R}{2}$ is placed in magnetic field region, then emf induced in the ring is $\frac{\pi\alpha R^2}{4}$.

Which of the above statements are correct?

(1) a & b

(2) b & c

(3) a, b, c

(4) All of above

PEI270

26. Consider currents flowing simultaneously in two nearly coils having their axis on line. Then flux linked with one coil is

(1) Only due to self induction

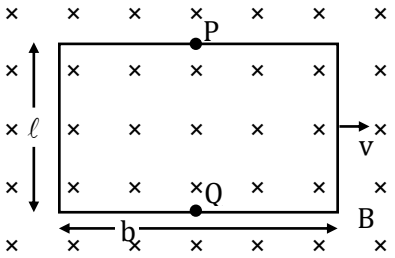
(2) Only due to mutual induction

(3) Both self and mutual induction

(4) Neither self or mutual induction

PEI271

27. Calculate the emf induced



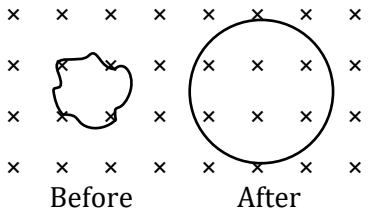
(a) in the loop (b) across PQ

(1) zero, vBℓ (2) zero, zero

(3) 2vBℓ, zero (4) vBℓ, vBℓ

PEI272

28. There is a uniform magnetic field directed perpendicular and into the plane of paper. An irregular shaped conducting loop is slowly changing into a circular loop. Then



(1) current is induced in ACW dir.

(2) current is induced in CW dir.

(3) no current is induced in loop.

(4) None of the above.

PEI273

29. Lenz Law is based on

(1) Conservation of charge

(2) Conservation of energy

(3) Both (1) & (2)

(4) None of the above

PEI274

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	4	3	1	2	2	2	3	4	4	3	2	2	3	3	3
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	
Answer	3	2	3	2	2	1	3	2	2	4	3	1	1	2	

Electromagnetic Waves

1. Displacement current is due to -
(1) Conduction of charge
(2) Variable electric field
(3) Variable magnetic field
(4) None of above

PEW083

2. In electromagnetic wave -
(1) Constant E & B are there
(2) Oscillating E produces constant B
(3) Oscillating B produces constant E
(4) Oscillating E & B regenerate each other

PEW084

3. About 5% of the power of a 100W light bulb is converted to visible radiation. Then average intensity of visible radiation at a distance 1m from bulb is
(1) 0.04 W/m^2 (2) 0.4 W/m^2
(3) 0.02 W/m^2 (4) 0.2 W/m^2

PEW085

4. Instantaneous displacement current of 2A is setup in the space between two parallel plates of 1μF capacitor. Then rate of change of potential difference across capacitor plates is
(1) 3×10^6 V/s (2) 4×10^6 V/s
(3) 2×10^6 V/s (4) None of these

PEW086

5. Ampere - Maxwell law is -
(1) $\oint \vec{B}.d\vec{\ell} = \mu_0 I_C + \mu_0 \in_0 \frac{d\phi_E}{dt}$
(2) $\oint \vec{B}.d\vec{\ell} = \mu_0 I_C + \in_0 \frac{d\phi_E}{dt}$
(3) $\oint \vec{B}.d\vec{\ell} = \in_0 \mu_0 I_C + \in_0 \frac{d\phi_E}{dt}$
(4) None of these

PEW087

6. Snakes can detect which of the following electro magnetic waves?
(1) X-rays (2) Gamma rays
(3) Ultraviolet rays (4) Infrared rays

PEW088

7. EMW exerts radiation pressure because -
(1) It can travel in vacuum
(2) E and B are mutually perpendicular
(3) It carries momentum
(4) None of these

PEW089

8. Which band of radio waves is used for communication in cellular phones -
(1) Very high frequency (VHF)
(2) Ultra high frequency (UHF)
(3) Super high frequency (SHF)
(4) None of these

PEW090

9. In an electromagnetic wave, the average energy density associated with magnetic field is
(1) $\frac{Li_0^2}{2}$ (2) $\frac{B_0^{\ 2}}{4\mu_0}$ (3) $\frac{\mu_0 B^2}{2}$ (4) $\frac{\mu_0}{2B^2}$

PEW091

10. Maxwell noticed inconsistency in -
(1) Gauss law in electrostates
(2) Gauss law in magnetism
(3) Ampere's circuital law
(4) Faraday's law

PEW092

11. Direction of wave vector K is -
(1) Along wave propagation
(2) Normal to wave propagation
(3) Along electric field
(4) Along magnetic field

PEW093

12. Which experiment marked the beginning of the field of communication using EMW.
(1) Hertz experiment
(2) Marconi experiment
(3) Devisson Germer Experiment
(4) None of these

PEW094

13. Electromagnetic wave can travel in -
(1) Vacuum only
(2) Medium only
(3) Vacuum and medium both
(4) None of these

PEW095

14. Velocity of light -
(1) Always constant
(2) Increases when medium changes
(3) Decreases when medium changes
(4) Depends on electric and magnetic properties of medium

PEW096

15. **Statement-1** : According to Ampere Maxwell law, $\oint \vec{B}\cdot\vec{dl} = \mu_0 i = \mu_0 (i_c + i_d)$ where i_c is conduction current and i_d is displacement current
Statement-2 : Since $i_c = i_d$, $i = 2i_c = 2i_d$
(1) Statement-1 is correct; statement 2 is incorrect
(2) Both statement-1 and statement 2 are correct
(3) Statement-1 is incorrect; statement 2 is correct
(4) Both Statement-1 and statement 2 are incorrect

PEW097

16. If the frequency of oscillation of charge is 10^{10}Hz then frequency of EMW is
(1) 2×10^{10} Hz (2) 10^{10} Hz
(3) $\frac{10^{10}}{2}$ Hz (4) None of these

PEW098

17. For an electromagnetic wave
$E_y = \left(120\frac{N}{C}\right)\sin[kx+\omega t]$
then direction of propagation of EMW is
(1) +x axis (2) –x axis
(3) +y axis (4) –y axis

PEW099

18. For an electromagnetic wave
B_Z = (10nT) sin [kx – ωt] then direction of propagation of EMW is
(1) –x axis (2) +x axis
(3) +z axis (4) –y axis

PEW100

19. For an electromagnetic wave in vaccum
$E_y = E_0\sin(kx - \omega t)$; $B_z = B_0 \sin(kx - \omega t)$
Then speed of EM wave is
(1) $\frac{E_0}{B_0}$ (2) $\frac{\omega}{k}$
(3) Both (1) & (2) (4) None of these

PEW101

20. Refractive index of a medium having relative permeability and relative permittivity as μ_r and ε_r respectively is: -
(1) $\frac{1}{\sqrt{\mu_r\varepsilon_r}}$ (2) $\sqrt{\mu_r\varepsilon_r}$
(3) $\frac{\mu_r}{\varepsilon_r}$ (4) None of these

PEW102

21. If the intensity of electromagnetic wave is I then part of intensity provided by the electric field is
(1) $\frac{I}{4}$ (2) $\frac{I}{2}$
(3) I (4) None of these

PEW103

22. Light with an energy flux of 18 W/cm^2 falls on a non reflecting surface at normal incidence. If the surface has an area of 20 cm^2, find the average force exerted on the surface during a 30 minute time span.
(1) 1.2×10^{-6}N (2) 1.2×10^{-5}N
(3) 2.4×10^{-6}N (4) 2.4×10^{-5}N

PEW104

23. When a radiation of an intensity I is incident on a perfectly reflective surface, the radiation pressure P exerted on surface is
(1) $\frac{I}{C}$ (2) $\frac{2I}{C}$
(3) $\frac{I}{2C}$ (4) None of these

PEW105

24. When a radiation of an intensity I is incident normally on a surface which reflects 70% and absorbs 30% of radiation, the radiation pressure P exerted on surface is

(1) $1\cdot1\frac{I}{c}$ (2) $1\cdot4\frac{I}{c}$

(3) $0\cdot3\frac{I}{c}$ (4) $1\cdot7\frac{I}{c}$

PEW106

25. The ratio of average energy density of electric field part of EMW to the average energy density of electromagnetic wave is

(1) 1 : 1 (2) 2 : 1

(3) 1 : 2 (4) 1 : 4

PEW107

26. **Assertion: -** Infrared radiation plays an important role in maintaining the average temperature of earth.

Reason: - Infrared radiations are sometimes referred to as heat waves.

(1) Both assertion and reason are true and reason is the correct explanation of assertion.

(2) Both assertion and reason are true and reason is not the correct explanation of assertion

(3) Assertion is true but reason is false

(4) Both assertion and reason are false

PEW108

27. X-rays, gamma rays and microwaves travelling in vaccum have

(1) Same wavelength but different velocities

(2) Same frequency but different velocities

(3) Same velocity but different wavelengths

(4) Same velocity and same frequency

PEW109

28. A :- Wavelength of microwaves is greater than that of UV rays

B :- Wavelength of infrared rays is lesser than that of UV rays

C :- Wavelength of microwaves is lesser than that of infrared rays

D :- Gamma Rays has shortest wavelength in the electromagnetic spectrum

Choose correct option from the given options

(1) A & B are true

(2) B & C are true

(3) C & D are true

(4) A & D are true

PEW110

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	2	4	2	3	1	4	3	2	2	3	1	2	3	4	1
Question	16	17	18	19	20	21	22	23	24	25	26	27	28		
Answer	2	2	2	3	2	2	1	2	4	3	2	3	4		

Ray Optics and Optical Instruments

1. If an object is placed in front of a concave mirror and the lower half of the mirror is covered with opaque material then what happens
(1) lower half of the object will be seen as image
(2) upper half of the object will be seen as image
(3) No change in the image will taken place
(4) image will be completely formed with less intensity

PRI303

2. Optical fibre consists of core and cladding whose refractive index are related as :-
(1) $\mu_{core} = \mu_{clading}$
(2) $\mu_{core} > \mu_{clading}$
(3) $\mu_{core} < \mu_{clading}$
(4) No relation between them

PRI304

3. While moving in a bus or a car during a hot summer day, a distant patch on road appears to be wet. It is due to :-
(1) Reflection
(2) Total internal reflection
(3) Scattering
(4) Dispersion

PRI305

4. Modern telescopes are designed in such a way that there is no chromatic aberration present. It is used by having :-
(1) Convex lens
(2) Concave mirror
(3) Both concave mirror and convex lens
(4) Prism

PRI306

5. Laws of reflection are strictly valid for :-
(1) Plane surfaces
(2) Rough surfaces
(3) All types of surfaces
(4) None of these

PRI307

6. In telescope, which mirror is used as objective instead of convex lens :-
(1) Parabolic concave mirror
(2) Plane mirror
(3) Convex mirror
(4) Mirror can't be used

PRI308

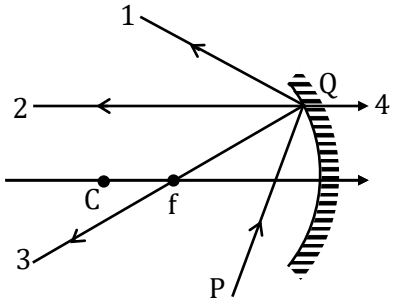
7. Relation between paraxial rays focus (f_p) & marginal ray's focus (f_m) in case of a spherical mirror: -
(1) $f_p = f_m$ (2) $f_p > f_m$
(3) $f_m > f_p$ (4) None of these

PRI309

8. In vacuum, all colours :-
(1) have same speed
(2) have different-different speed
(3) do not move
(4) absorb all colours

PRI310

9. The direction of ray of light incident on a concave mirror is shown by PQ while direction in which the ray would travel after reflection is shown by four rays, marked 1,2,3 and 4 which of the four rays correctly shows the direction of reflected ray?



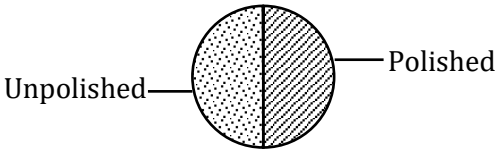
(1) 1 (2) 2 (3) 3 (4) 4

PRI311

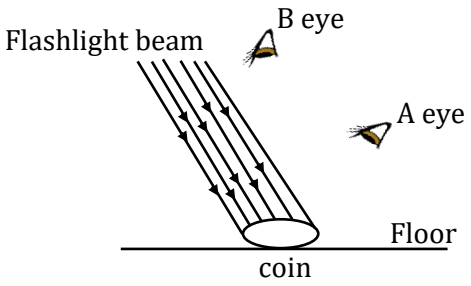
10. An air bubble is formed inside water. It act as a :-
(1) convex mirror (2) converging lens
(3) diverging lens (4) plane mirror

PRI312

11. Consider the metal coin in figure. The right half is polished, while left half is unpolished.



In a dark room, this coin is placed face up on the floor and illuminated with a thin-beamed flashlight, as shown in figure. A and B look at the coin from the position as indicated in figure.



According to A, which half of the coin (if either) is brighter?

(1) The polished half

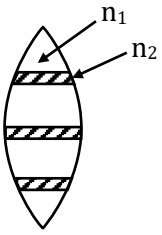
(2) The unpolished half

(3) Both halves look equally bright

(4) Both halves look completely dark

PRI313

12. Shown in the figure is a bi-convex lens made of different transparent strips of two different refractive indices. If the parallel rays of light is incident on the lens then select INCORRECT statement: -



(1) rays will intersect before focussing

(2) rays will be focussed at two different points

(3) the lens may have four focal points different from each other.

(4) None of the these

PRI314

13. The apparent depth of a tank filled with water: -

(1) Will be more for oblique viewing

(2) Will be more for normal viewing

(3) Is independent of the angle of viewing

(4) None of above

PRI315

14. The power of a biconvex lens is 10 dioptre and the radius of curvature of each surface is 10 cm. Then the refractive index of the material of the lens is

(1) $\frac{3}{2}$ (2) $\frac{4}{3}$ (3) $\frac{9}{8}$ (4) $\frac{5}{3}$

PRI316

15. An eye specialist prescribes spectacles having a combination of a convex lens of focal length 40 cm in contact with a concave lens of focal length 25 cm. The power of this lens combination is: -

(1) +1.5 D (2) –1.5 D

(3) + 6.67 D (4) – 6.67 D

PRI317

16. For a glass prism $\left(\mu=\sqrt{3}\right)$ the angle of minimum deviation is equal to the angle of the prism. The angle of the prism is :-

(1) 45° (2) 30° (3) 90° (4) 60°

PRI318

17. In a compound microscope, the focal lengths of two lenses are 1.5 cm and 6.25 cm. If an object is placed at 2 cm from objective and the final image is formed at 25 cm from eye lens, the distance between the two lenses is

(1) 6.00 cm (2) 7.75 cm

(3) 9.25 cm (4) 11.0 cm

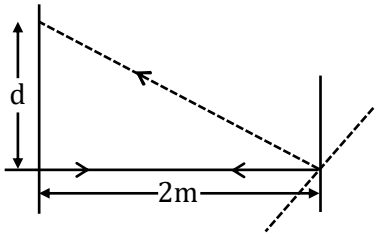
PRI319

18. A reflecting telescope has a large mirror for its objective with radius of curvature equal to 80 cm. The magnifying power of this telescope if eye piece used has a focal length of 1.6 cm is: -

(1) 100 (2) 50 (3) 25 (4) 5

PRI320

19. Light incident normally on a plane mirror attached to a galvanometer coil reflects backward as shown in figure. A current in the coil produces a deflection of 3.5° of the mirror. The displacement of the reflected spot of light on a screen placed 2m away is: -



(1) 27.5 m (2) 48.9 cm

(3) 24.5 cm (4) 12.2 m

PRI321

20. Paraxial rays in geometrical optics are those rays which are
(1) parallel to principal axis making large angle of incidence
(2) non-parallel to principal axis
(3) parallel to principal axis making small angle of incidence
(4) None

PRI322

21. A magnifying lens is used, as the object to be viewed can be brought closer to the eye than the normal near point. This :-
(1) A larger angle to be subtended by the object at the eye and hence, viewed in greater detail.
(2) The formation of real erect image.
(3) Increase in the field of view
(4) Infinite magnification at the near point

PRI323

22. If a Ray is incident on plane surface, than after reflection
(1) Frequency changes
(2) Wave changes
(3) Speed changes
(4) Frequency, wavelength and speed remains same

PRI324

23. If a incident ray is diverging nature, than object is: -
(1) Real
(2) Virtual
(3) Both real & virtual are possible
(4) None of the above

PRI325

24. An unnumbered wall clock shows time 10:25:37, where Ist term represents hours, IInd represents minutes and the last term represents seconds. What time will its image in a plane mirror shows?
(1) 01:34:33 (2) 01:34:23
(3) 01:33:33 (4) 01:33:23

PRI326

25. If a mirror rotated by 'θ' angle in clockwise direction than reflected ray rotates by angle
(1) θ, clockwise (2) 2θ, anticlockwise
(3) 2θ, clockwise (4) θ, anticlockwise

PRI327

26. For a simple microscope, the angular size of object and angular size of images are
(1) Unequal
(2) Equal
(3) Angular size of object is more
(4) Angular size of image is more

PRI328

27. For a optically rarer medium, speed of light will be: -
(1) more (2) less
(3) same as before (4) decreased

PRI329

28. While applying lens formula correct statement is
(1) Rays should be marginal
(2) Lenses should be thick
(3) Medium on both side of the lens should be same
(4) None of the above

PRI330

29. When light ray enter in a glass slab from water their wavelength: -
(1) Remains same (2) Decreases
(3) Increases (4) Either (1) or (2)

PRI331

30. **Assertion (A): -** Mass density of turpentine is less than that of water but its optical density is higher.
Reason (R): - Optical density and mass density are related to each other.
(1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.
(2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.
(3) **(A)** is true but **(R)** is false.
(4) **(A)** is false but **(R)** is true.

PRI332

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31. When a ray of light entering glass from air, it is partially reflected and partially refracted. This reflection is called: -

(1) Total inter reflection

(2) Refraction

(3) Reflection

(4) Internal Reflection

PRI333

32. In convex lens of focal length F, minimum distance between an object and its real image must be: -

(1) 3F

(2) 4F

(3) $\frac{3}{2}$F

(4) 2F

PRI334

33. The tangent of angle by which it converges or diverges a beam of light falling at unit distance from optical centre is known as: -

(1) Focus (2) Radius

(3) Power (4) None of these

PRI335

34. The distance measured in the same direction as the incident light rays are moving is taken as

(1) Negative (2) Positive

(3) Both (1) & (2) (4) None of the above

PRI336

35. When a concave mirror is used to form a image of a real object, then which magnification is not possible: -

(1) m = –1 (2) $m = -\frac{1}{3}$

(3) m = +2 (4) $m = +\frac{1}{2}$

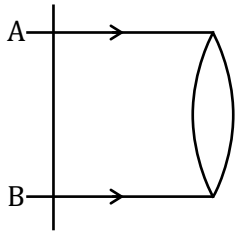
PRI337

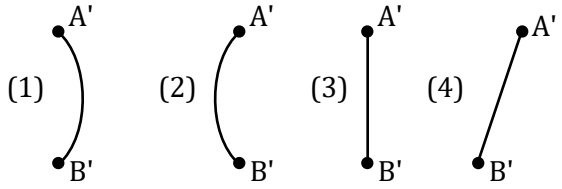
ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	4	2	2	2	3	1	2	1	1	3	1	4	2	1	2
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	4	4	3	3	3	1	4	1	2	3	2	1	3	2	3
Question	31	32	33	34	35										
Answer	4	2	3	2	4										

Wave Optics

1. If AB is incident wavefront. Then refracted wavefront (just after transmission) is :-





PWO235

2. Two source are said to be coherent when :-

(1) Phase difference between waves produced by them is a function of time

(2) Phase difference between waves produced by them is $\pi/2$

(3) Phase difference between waves produced by them is 0

(4) Phase difference between them remains constant with time

PWO236

3. In the phenomenon of interference, energy is :-

(1) Destroyed at destructive interference

(2) Created at constructive interference

(3) Conserved but it is redistributed

(4) Same at all points

PWO237

4. In young's double-slit experiment the central bright fringe can be identified.

(1) as it has greater intensity than the other bright fringe

(2) as it is wider than the other bright fringes

(3) as it is narrower than the other bright fringe

(4) by using white light instead of monochromatic light

PWO238

5. Intensity of two interfering coherent sources is 9 units and 4 units respectively. What is the maximum intensity in the interference pattern?

(1) 25 unit (2) 13 unit

(3) 1 unit (4) 5 unit

PWO239

6. The slits in a Y.D.S.E. have equal widths and the source is placed symmetrically relative to the slits. The intensity at the central fringe is $\frac{I_0}{4}$. If one of the slits is closed, the intensity at this point will be :-

(1) I_0 (2) $\frac{I_0}{4}$ (3) $\frac{I_0}{16}$ (4) $\frac{I_0}{32}$

PWO240

7. A light source of 6000 Å wave length produces a single slit diffraction. The first minima in diffraction pattern is seen, at a distance of 6 mm from central maxima. The distance between screen and slit is 2 metre. The width of slit in mm will be :

(1) 0.1 (2) 0.4

(3) 0.2 (4) 2

PWO241

8. The intensity at the maximum in a Young's double slit experiment is I_0. Distance between two slits is d = 5λ, where λ is the wavelength of light used in the experiment. What will be the intensity in front of one of the slits on the screen placed at a distance D = 10 d?

(1) I_0 (2) $\frac{I_0}{4}$

(3) $\frac{3}{4}I_0$ (4) $\frac{I_0}{2}$

PWO242

9. If the amplitude ratio of two sources producing interference is 3 : 5, the ratio of intensities at maxima and minima is :

(1) 25 : 16 (2) 5 : 3

(3) 16 : 1 (4) 25 : 9

PWO243

10. At the first minimum adjacent to the central maximum of a single-slit diffraction pattern the phase difference between the Huygen's wavelet from the edges of the slit is :-

(1) $\frac{\pi}{8}$ radian (2) $\frac{\pi}{4}$ radian

(3) $\frac{\pi}{2}$ radian (4) 2π radian

PWO244

11. When light waves suffer reflection at the interface between air and glass, the change of phase of the reflected wave is equal to :-

(1) 0 (2) $\pi/2$ (3) π (4) 2π

PWO245

12. The interference pattern is obtained with two coherent light sources of intensity ratio 9. In the interference pattern, the ratio $\frac{I_{max}-I_{min}}{I_{max}+I_{min}}$ will be : -

(1) 5/7 (2) 3/5 (3) 1/2 (4) 4/3

PWO246

13. Angular width of the central maxima in the Fraunhofer diffraction for λ = 6000 Å is θ_0. When the same slit is illuminated by another monochromatic light, the angular width decreases by 30%. The wavelength of this light is,

(1) 1800 Å (2) 4200 Å

(3) 6000 Å (4) 420 Å

PWO247

14. The two coherent sources with intensity ratio β produce interference. The fringe visibility will be:

(1) $\frac{2\sqrt{\beta}}{1+\beta}$ (2) 2β

(3) $\frac{2}{1+\beta}$ (4) $\frac{\sqrt{\beta}}{1+\beta}$

PWO248

15. A thin sheet of glass (μ = 1.5) of thickness 6 microns introduced in the path of one of interfering beams in a double slit experiment shifts the central fringe to a position previously occupied by fifth dark fringe. Then the wavelength of light used is :-

(1) 6600 Å (2) 3000 Å

(3) 4500 Å (4) 7500 Å

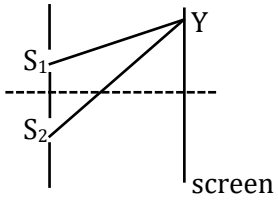
PWO249

16. Two waves represented by $y_1 = 4 \sin \omega t$ and $y_2 = 3 \sin\left(\omega t + \frac{\pi}{3}\right)$ interfere at a point. Find out the approximate amplitude of the resulting wave.

(1) 3 (2) 6 (3) 5 (4) 8

PWO250

17. Figure shows path of white light rays which leave in phase from two small sources S_1 and S_2 and travel to a point Y on the screen. The path difference $S_2Y - S_1Y = 11.4 \times 10^{-7}$ m.
What wavelength of light gives brightest illumination at Y?



(1) 3.8×10^{-7} m (2) 5.2×10^{-7} m

(3) 7.6×10^{-7} m (4) 6.4×10^{-7} m

PWO251

18. Light of wavelength 6000 Å is incident on a thin glass plate such that angle of refraction into the plate of 60°. If the smallest thickness of plate which makes it appear dark by the reflection is 4×10^{-7} m. What is the refractive index of glass plate :

(1) $\frac{5}{3}$ (2) $\frac{3}{2}$ (3) $\frac{4}{3}$ (4) 2

PWO252

19. Two beams, A and B, of plane polarized light with mutually perpendicular planes of polarization are seen through a polaroid. From the position when the beam A has maximum intensity (and beam B has zero intensity), a rotation of polaroid through 30° makes the two beams appear equally bright If the initial intensities of the two beams are I_A and I_B respectively, then $\frac{I_A}{I_B}$ equals :

(1) 1 (2) $\frac{I}{3}$ (3) 3 (4) $\frac{3}{2}$

PWO253

20. The phase difference ($\Delta\phi$) between 2 superimposing waves for bright fringe (maxima) is: -

(1) $\Delta\phi = n\pi$ n = 1,2,3,4,5

(2) $\Delta\phi = 2n\pi$ n = 0,2,4,6

(3) $\Delta\phi = 2n\pi$ n = 0,1,2,3,4,5

(4) $\Delta\phi = \frac{3n\pi}{2}$ n = 1,2,3,4,5

PWO254

21. Colourful patter seen on a CD is due to the phenomenon: -

(1) Polarisation (2) Interference

(3) Scattering (4) Diffraction

PWO255

22. **Assertion: -** Radio waves can be polarized.

Reason: - Radio waves are longitudinal waves.

(1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.

(2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.

(3) **(A)** is true but **(R)** is false.

(4) **(A)** is false but **(R)** is true.

PWO256

23. If fringe width is 0.4mm, then distance between the fifth bright and third dark band on the same side is: -

(1) 0.6mm (2) 0.4mm

(3) 1.0mm (4) 0.8mm

PWO257

24. If YDSE is performed in water: -

(1) Angular fringe width will not change

(2) Linear fringe width will not change

(3) Angular fringe width will increase

(4) Linear fringe width will decrease

PWO258

25. Plane polarized light can be obtained by using: -

(1) Nicol prism

(2) Tourmaline crystal

(3) Convex lens

(4) Both (1) & (2)

PWO259

26. Two waves of equal amplitude 'A' and same wavelength 'λ' from 2 coherent sources (S_1 & S_2) interfere at a point 'P' such that $S_2P - S_1P = \lambda/3$. The amplitude at point 'P' will be: -

(1) $\frac{\sqrt{3}A}{2}$ (2) $\frac{A}{2}$

(3) A (4) 2A

PWO260

27. When a polarised beam is passed through a polaroid, the emergent amplitude of beam will be: - (If initial amplitude is E)

(1) E sinθ (2) E cosθ

(3) E tanθ (4) E cot θ

PWO261

28. Polaroids can be used to control intensity in: -

(1) Sunglasses (2) 3D movie camera

(3) Windowpanes (4) All the above

PWO262

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	2	4	3	4	1	3	3	4	3	4	3	2	2	1	1
Question	16	17	18	19	20	21	22	23	24	25	26	27	28		
Answer	2	1	2	2	3	4	3	3	4	4	3	2	4		

Modern Physics

1. With regard to neutrinos, pick the incorrect statement :
(1) They are generated during β-decays
(2) They have strong interactions with other particles
(3) They are neutral particles
(4) They are hard to detect

PMD329

2. Heavy stable nucleus has more neutrons than protons. This is because of the fact that
(1) neutrons are heavier than protons.
(2) electrostatic force between protons are repulsive.
(3) neutrons decay into protons through beta decay.
(4) nuclear forces between neutrons are weaker than that between protons.

PMD330

3. The wave-length of the matter-wave depends on :
(1) Mass (2) Velocity
(3) Momentum (4) All of these

PMD331

4. The photo electrons emitted from a surface of sodium metal are such that they:
(1) All are of the same frequency
(2) Have the same KE
(3) Have the same de-Broglie wave-length
(4) Have their speed varying from zero to a certain maximum

PMD332

5. In a photoelectric experiment, if stopping potential is applied, the photo current becomes zero. This means that:
(1) The emission of photoelectron is stopped
(2) The photoelectrons are emitted but are accumulated near the emitter metal surface.
(3) The photoelectrons are dispersed from the emitter surface
(4) None of these

PMD333

6. If the frequency of light in a PEE is doubled then maximum KE of photoelectron:
(1) Doubled
(2) Be halved
(3) Become more than double
(4) Become less than double

PMD334

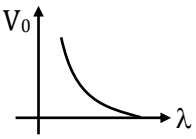
7. Stopping potential for photoelectrons:
(1) Does not depend on the frequency of incident light
(2) Does not depend on the nature of cathode material
(3) Depends on both the frequency of incident light and nature of cathode material
(4) Depends on intensity of incident light

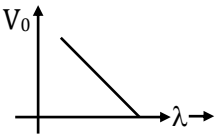
PMD335

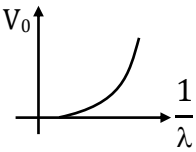
8. If an electron and a photon propagate in form of waves having the same wavelength, it implies that they can have the same: -
(1) Energy
(2) Momentum
(3) Velocity
(4) Angular momentum

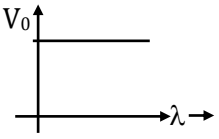
PMD336

9. For photo-electric effect,
$\lambda \rightarrow$ incident photon wavelength
$V_0 \rightarrow$ stopping potential.

(1) 

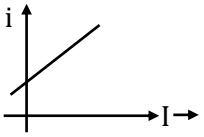
(2) 

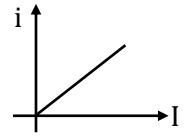
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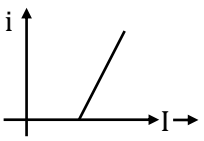
(4) 

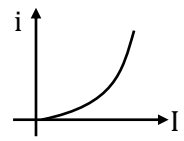
PMD337

10. The graph between intensity (I) of light falling on a metallic plate with current (i) generated is:

(1) 

(2) 

(3) 

(4) 

PMD338

11. The equation E = pc is valid:
(1) For an electron as well as for photon
(2) For an electron but not for a photon
(3) For a photon but not for electron
(4) Neither for an electron nor a photon

PMD339

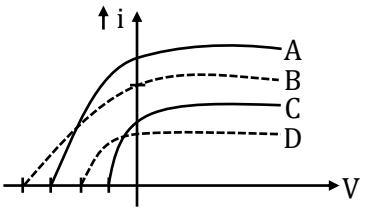
12. In photoelectric effect, if a weak intensity radiation instead of strong intensity of suitable frequency is used then:
(1) Photoelectric effect will get delayed
(2) Photoelectric effect will not take place
(3) Maximum KE will decrease (for photo electron)
(4) Saturation current will decrease

PMD340

13. When a nucleus in an atom undergoes a radioactive decay, the electronic energy levels of the atom
(1) do not change for any type of radioactivity.
(2) change for α and β radioactivity but not for γ radioactivity.
(3) change for α radioactivity but not for others.
(4) change for β radioactivity but not for others.

PMD341

14.



The figure shows the result of an experiment involving photoelectric effect. The graph A, B, C and D relate to a light beam having different wavelength. Select correct:
(a) Beam B has a highest frequency
(b) Beam C has longest wavelength
(c) Beam A has highest rate of photoelectric emission
(d) Highest momentum is possesed by some electrons emitted by B
(1) a & b only (2) a, b & c
(3) a, b, c & d (4) a & c only

PMD342

15. The photocurrent in experiment of photo electric effect increases if:
(1) intensity of source is increased
(2) exposure time is increased
(3) exposure time is decreased
(4) intensity of source is decreased

PMD343

16. Energy conversion in photoelectric cell takes place from :-
(1) Chemical to electrical
(2) Magnetic to electrical
(3) Optical to electrical
(4) Mechanical to electrical

PMD344

17. Atomic binding energy of elements A and B are E_a and E_b respectively. Three atoms of B form one atom of A. In this fusion process E energy is released, the relation between E_a, E_b and E :-
(1) $E_a + E = 3E$ (2) $E_a = 3E_b$
(3) $E_a - E = 3E_b$ (4) $E_a + 3E_b + E = 0$

PMD345

18. An radioactive atom ($_ZX^A$) undergoes α-decay then ratio of energy of α-particle and daughter nuclei increases with increase in :-
(1) A^2 (2) A^0 (3) A (4) A^{-1}

PMD346

19. Fusion reaction takes place at high temperature because :-
(1) Molecules break up at high temperature
(2) Nuclei break up at high temperature
(3) Atoms get ionised at high temperature
(4) Kinetic energy is high enough to overcome the coulomb repulsion between nuclei

PMD347

20. A nuclear fusion reaction is given below :
$_1H^2 + {}_1H^2 \rightarrow {}_2He^3 + n + 3.2$ MeV
How much energy will be generated when 2 kg of deuterons are fused :- (approx)
(1) 10^{30} eV (2) 5×10^{23} MeV
(3) 10^{22} MeV (4) 10^{33} eV

PMD348

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21. Outside a nucleus :-
(1) Neutron is stable
(2) Proton & neutron both are stable
(3) Neutron is unstable
(4) Neither neutron nor proton is stable
PMD349

22. Lenard observed that no electrons are emitted when frequency of light is less than a certain minimum frequency. This minimum frequency depends on:
(1) Potential difference of emitter and collector plates
(2) Distance between collector and the emitter plate
(3) Size (area) of the emitter plate
(4) Material of the emitter plate
PMD350

23. Macroscopic objects in our daily life do not show wave like properties:
(1) de-Broglie's hypothesis is not true
(2) de-Broglie's hypothesis is true only for electron
(3) de-Broglie's hypothesis is true only for photons
(4) Wavelength associated with macroscopic object is small
PMD351

24. Which of the following statement are true:
(a) Dual nature of matter is evident in de-Broglie equation $\lambda = \frac{h}{p}$.
(b) Wave nature of macroscopic particles are not visible in daily life
(c) $\lambda = \frac{h}{p}$ is not satisfied by a photon
(1) (a, c) (2) (a, b)
(3) (b, c) (4) (a, b, c)
PMD352

25. The electron are emitted in the photoelectric effect from a metal surface: -
(1) Only if the frequency of the incident radiation is above a certain threshold value
(2) Only if the temperature of the surface is high
(3) At a rate that is independent of the nature of the metal
(4) With a maximum velocity proportional to the frequency of the incident radiation
PMD353

26. Nuclear forces acting between two nucleons inside nucleus, then incorrect statement is :
(1) Has saturation property
(2) Is short ranged which explains the constancy is BE/A in the range $30 < A < 170$.
(3) Has zero associated potential energy associated when distance between nucleons is 0.8 fm.
(4) Is the strongest force inside nucleus
PMD354

27. In a thermonuclear fusion, which statement is incorrect :
(1) Energy is provided by raising the temperature
(2) Energy must be provided to overcome columbic repulsion between approaching positively charged nuclei.
(3) Is responsible for the formation of red giant star
(4) Required temperature is of the order of 10^8 K.
PMD355

28. Which of the following statement is incorrect for a nuclear reaction
(1) If nuclei with less total binding energy transform to nuclei with greater binding energy, there will be a net energy release.
(2) Nuclear reactions are not balanced in the same way as chemical reactions
(3) Both the number of protons and number of neutrons are conserved, but the total mass is not conserved.
(4) In β^-decay a neutrino is generated, while is β^+ decay an anti-neutrino is generated
PMD356

29. Pick the incorrect statement :
(1) Fission of 1kg of Uranium generates equal energy as burning of 1kg of coal.
(2) During fission of $_{92}U^{235}$, different pairs of intermediate mass fragments can be obtained
(3) The peaks at nuclides like $_{2}He^{4}$, $_{8}O^{16}$ is an evidence of shell-like structure inside nuclei.
(4) Decay of unstable $_{6}C^{14}$ isotope into stable $_{7}N^{14}$ is the principle of carbon dating.

PMD357

30. The wave nature of light shows up in the phenomenon of .....?
(1) Interference (2) Diffraction
(3) Polarisation (4) All of the above

PMD358

31. The minimum energy needed by an electron to come out from a metal surface is called?
(1) Kinetic energy (2) Work function
(3) Potential energy (4) None of these

PMD359

32. Photoelectric current depends upon?
(1) The intensity of incident light
(2) The potential difference applied between the two electrodes
(3) The nature of the emitter material
(4) All (1), (2) and (3)

PMD360

33. In a experiment of photo electric effect, the slope of the cut-off voltage versus frequency of incident light is found to be 4.12×10^{-15}V-S. Calculate the value of plank constant?
(1) 6.59×10^{-34} J-s (2) 6.34×10^{-34} J-s
(3) 6.02×10^{-34} J-s (4) None

PMD361

34. A 12.5 eV electron beam is used to bombard gaseous hydrogen at room temperature. What series of wavelengths will be emitted?
(1) Lyman series: 103 nm and 122 nm
(2) Balmer series: 656 nm
(3) Both (1) & (2)
(4) None

PMD362

35. At room temperature, most of the hydrogen atoms are in .....?
(1) Excited state (2) Ground state
(3) First excited state (4) None of these

PMD363

36. Given the mass of iron nucleus as 55.85u and A = 56, find the nuclear density?
(1) 2.29×10^{17} kg m^{-3} (2) 4.58×10^{17} kg m^{-3}
(3) 1.14×10^{17} kg m^{-3} (4) 0.57×10^{17} kg m^{-3}

PMD364

37. The radius of nucleus is smaller than the radius of an atom by a factor of .....?
(1) 10^4 (2) 10^2 (3) 1 (4) 10^8

PMD365

38. Obtain the binding energy (in MeV) of a nitrogen nucleus $\left(^{14}_{7}N\right)$, given $m\left(^{14}_{7}N\right) = 14.00307$ u
(1) 52.35 MeV (2) 104.7 MeV
(3) 209.4 MeV (4) None of these

PMD366

39. The radius of the first Bohr orbit is a_0. The nth orbit has a radius: -
(1) na_0 (2) a_0/n
(3) n^2a_0 (4) a_0/n^2

PMD367

40. How long can an electric lamp of 100W be kept glowing by fusion of 2.0 kg of deuterium? Take the fusion reaction as
$$^{2}_{1}H + ^{2}_{1}H \longrightarrow ^{3}_{2}He + n + 3.27\,MeV$$
(1) About 4.9×10^4 year
(2) About 9.8×10^4 year
(3) About 2.45×10^4 year
(4) About 1.64×10^4 year

PMD368

41. Slope between photoelectric current and intensity of light is : -
(1) Positive (2) Negative
(3) Zero (4) Infinite

PMD369

42. Photons are deflected by : -
(1) Electric field
(2) Magnetic field
(3) Both electric & magnetic field
(4) Neither electric nor magnetic field

PMD370

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43. In a photon-particle collision
(1) Total energy is conserved
(2) Momentum is conserved
(3) Both (1) & (2)
(4) Number of photons are conserved
PMD371

44. Charges of electrons and positron: -
(1) are equal in magnitude and same
(2) are equal in magnitude and opposite
(3) are unequal in magnitude and same
(4) are unequal in magnitude and opposite
PMD372

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	2	2	4	4	2	3	3	2	1	2	3	4	2	3	1
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	3	3	3	4	4	3	4	4	2	1	3	3	4	1	4
Question	31	32	33	34	35	36	37	38	39	40	41	42	43	44	
Answer	2	4	1	3	2	1	1	2	3	1	1	4	3	2	

Semiconductor Electronics

1. The depletion layer of a p-n junction :
(1) is of constant width irrespective of the bias
(2) acts like an insulating zone under reverse bias
(3) has a width that increases with an increases in forward bias
(4) is depleted of ions

PSE254

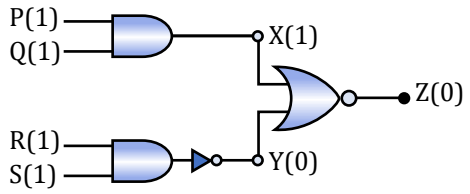
2. Which of the following statements is incorrect:
(1) The resistance of intrinsic semiconductor decrease with increase of temperature.
(2) Doping pure Si with trivalent impurities gives P-type semiconductors.
(3) The majority carriers in N-type semiconductors are holes.
(4) A PN-junction can act as a semiconductor diode.

PSE255

3. **Statement-1:** The number of free electrons in n-type semiconductor is higher than the number of free electrons in a pure silicon semiconductor.
Statement-2: The law of mass action is applicable only for n-type semiconductors.
(1) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
(2) Statement-1 is True, Statament-2 is True; Statement-2 is NOT a correct explanation for Statement-I.
(3) Statement-1 is True, Statement-2 is False.
(4) Statement-1 False, Statement-2 is True.

PSE256

4. The circuit diagram shows a logic combination with the states of output X, Y and Z given for inputs P, Q, R and S all at state 1. When inputs P and R change to state 0 with inputs Q and S still at 1, the states of outputs X, Y and Z change to :



(1) 1, 0, 0 (2) 1, 1, 1
(3) 0, 1, 0 (4) 0, 0, 1

PSE257

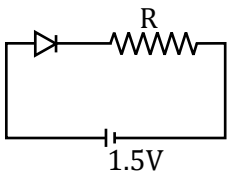
5. Energy band gap of silicon is 1.14 eV. Maximum wavelength at which silicon will start showing conductivity is?
(1) 10888 nm (2) 1088.8 nm
(3) 108.88 nm (4) 10.888 nm

PSE258

6. Doping of intrinsic semiconductor is done :
(1) To neutralize charge carriers
(2) To increase the concentration of majority charge carriers
(3) To make it neutral before disposal
(4) To carry out further purification

PSE259

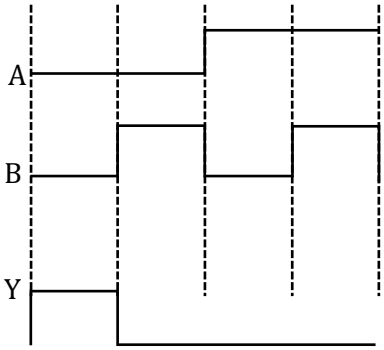
7. The diode used in a circuit shown in figure has constant voltage drop of 0.5V at all currents and maximum power rating of 100 mw. What should be value of resistance R connected in series with diode for obtaining maximum current



(1) 20Ω (2) 10Ω (3) 5Ω (4) 1.2Ω

PSE260

8. There is a gate with two inputs A and B, and one output Y. Using the input and output waveforms, identify the gate :



(1) NAND (2) NOR (3) OR (4) AND

PSE261

9. In a pure silicon ($n_i = 10^{16}$ /m^3) crystal at 300 K, 10^{21} atoms of phosphorus are added per cubic meter. The new hole concentration will be :-

(1) 10^{21} per m^3 (2) 10^{19} per m^3

(3) 10^{11} per m^3 (4) 10^{5} per m^3

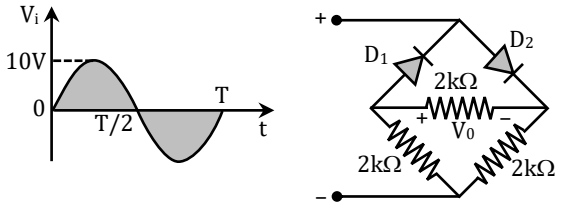
PSE262

10. Potential drop across a forward biased p-n junction diode is 0.7 V. If a battery of 4 V is applied, calculate the minimum resistance to be put in series, for the maximum current of 5 mA in the circuit.

(1) 660Ω (2) 350Ω (3) 475Ω (4) 500Ω

PSE263

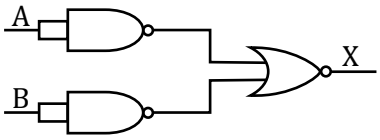
11. In the circuit shown in figure the maximum output voltage V_0 is



(1) 0 V (2) 5 V (3) 10 V (4) $\frac{5}{\sqrt{2}}$ V

PSE264

12. The truth table for the gate shown in the following figure will be



(1)

A	B	X
0	0	0
0	1	1
1	0	1
1	1	0

(2)

A	B	X
0	0	0
0	1	1
1	0	1
1	1	1

(3)

A	B	X
0	0	1
0	1	1
1	0	1
1	1	0

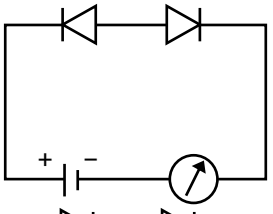
(4)

A	B	X
0	0	0
0	1	0
1	0	0
1	1	1

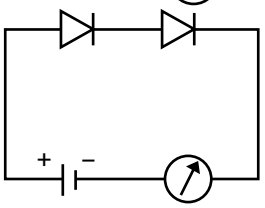
PSE265

13. Which circuit will not show current in ammeter :

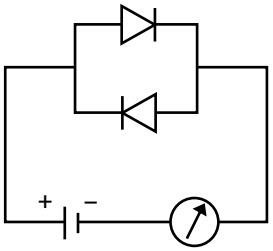
(1)



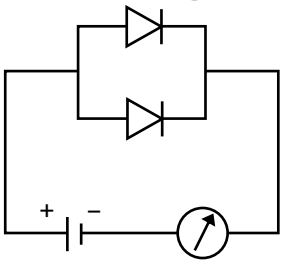
(2)



(3)

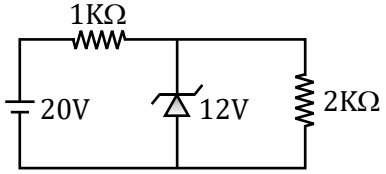


(4)



PSE266

14. In the given circuit the current through the zener diode is :-

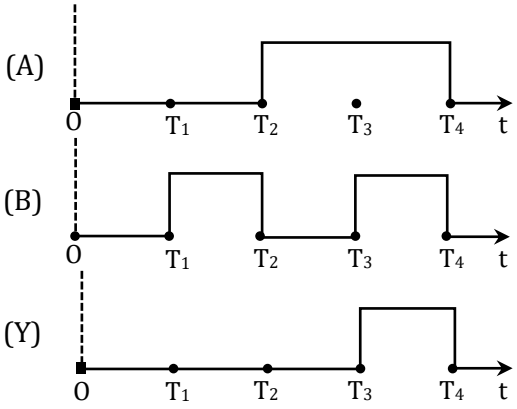


(1) 1 mA (2) 2 mA

(3) 6 mA (4) 6.67 mA

PSE267

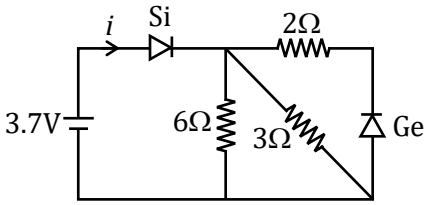
15. The given figure shows the wave forms for two inputs A and B and that for the output Y of a logic circuit. The logic circuit is



(1) An AND gate (2) An OR gate
(3) A NAND gate (4) An NOT gate

PSE268

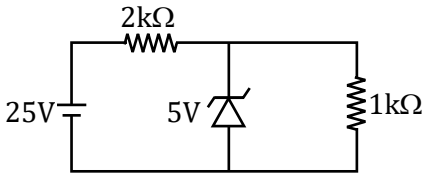
16. Find current 'i' :



(1) 2A (2) 3A (3) 1.5A (4) 0.5A

PSE269

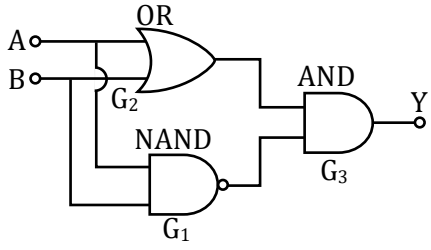
17. Find current through zener diode :-



(1) 5mA (2) 7.5 mA
(3) 2.5 mA (4) 12.5 mA

PSE270

18. The following configuration of gate is equivalent to



(1) NAND (2) XOR
(3) OR (4) None

PSE271

19. In an n-type silicon, which of the following statement is true:

(1) Electrons are majority carriers and trivalent atoms are the dopants.
(2) Electrons are minority carriers and pentavalent atoms are the dopants.
(3) Holes are minority carriers and pentavalent atoms are the dopants.
(4) Holes are majority carriers and trivalent atoms are the dopants.

PSE272

20. In an unbiased p-n junction, holes diffuse from the p-region to n-region because

(1) free electrons in the n-region attract them.
(2) they move across the junction by the potential difference.
(3) hole concentration in p-region is more as compared to n-region.
(4) All the above

PSE273

21. Carbon, silicon and germanium have four valence electrons each. These are characterised by valence and conduction bands separated by energy band gap respectively equal to $(E_g)_C$, $(E_g)_{Si}$ and $(E_g)_{Ge}$. Which of the following statements is true?

(1) $(E_g)_{Si} < (E_g)_{Ge} < (E_g)_C$ (2) $(E_g)_C < (E_g)_{Ge} > (E_g)_{Si}$
(3) $(E_g)_C > (E_g)_{Si} > (E_g)_{Ge}$ (4) $(E_g)_C = (E_g)_{Si} = (E_g)_{Ge}$

PSE274

22. When a forward bias is applied to a p-n junction, it

(1) raises the potential barrier.
(2) reduces the majority carrier current to zero.
(3) lowers the potential barrier.
(4) None of the above.

PSE275

23. What happens to the forbidden energy gap of a semiconductor with the fall of temperature?

(1) Decreases
(2) Increases
(3) Unchanged
(4) First increases and than decreases

PSE276

24. Semiconductors of both P-type and n-type are produced by: -
(1) ionic solids (2) covalent solids
(3) metallic solids (4) molecular solids

PSE277

25. A P-N junction diode allows current to pass only when it is .....?
(1) Reversed Biased.
(2) Forward Biased.
(3) Both Forward and Reversed Biased
(4) None of these

PSE278

26. In n-type semiconductors: -
(1) $n_e << n_h$ (2) $n_e >> n_h$
(3) $n_e = n_h$ (4) $n_e = n_h = n_i$

PSE279

27. Conductivity of an intrinsic semiconductor at room temperature is?
(1) very high (2) high
(3) very low (4) low

PSE280

28. Photo diode is operated in?
(1) Forward bias
(2) Reverse bias
(3) In both forward and Reverse bias
(4) None of these

PSE281

29. Find out the incorrect option?
(1) $\overline{A \cdot B} = \bar{A} + \bar{B}$ (2) $\overline{A + B} = \bar{A} \cdot \bar{B}$
(3) $A + A = 0$ (4) $A \cdot A = A$

PSE282

30. What is the necessity of doping?
(1) To increase the conductivity
(2) To decrease the conductivity
(3) To uneffact the conductivity
(4) None of these

PSE283

31. In a P-N junction under equilibrium :-
(1) $|I_{drift}| > |I_{diffusion}|$ (2) $|I_{drift}| < |I_{diffusion}|$
(3) $|I_{drift}| = |I_{diffusion}|$ (4) $|I_{drift}| \neq |I_{diffusion}|$

PSE284

32. Width of depletion region and doping is related as: -
(1) width $\propto$ doping
(2) width $\propto \frac{1}{\text{doping}}$
(3) width $\propto \sqrt{\text{doping}}$
(4) width $\propto \frac{1}{\sqrt{\text{doping}}}$

PSE285

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	2	3	3	3	2	2	3	2	3	1	2	4	1	2	1
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	3	1	2	3	3	3	3	2	2	2	2	3	2	3	1
Question	31	32													
Answer	3	4													